California ISO

Seams Workshop – Western Loading Relief process examples

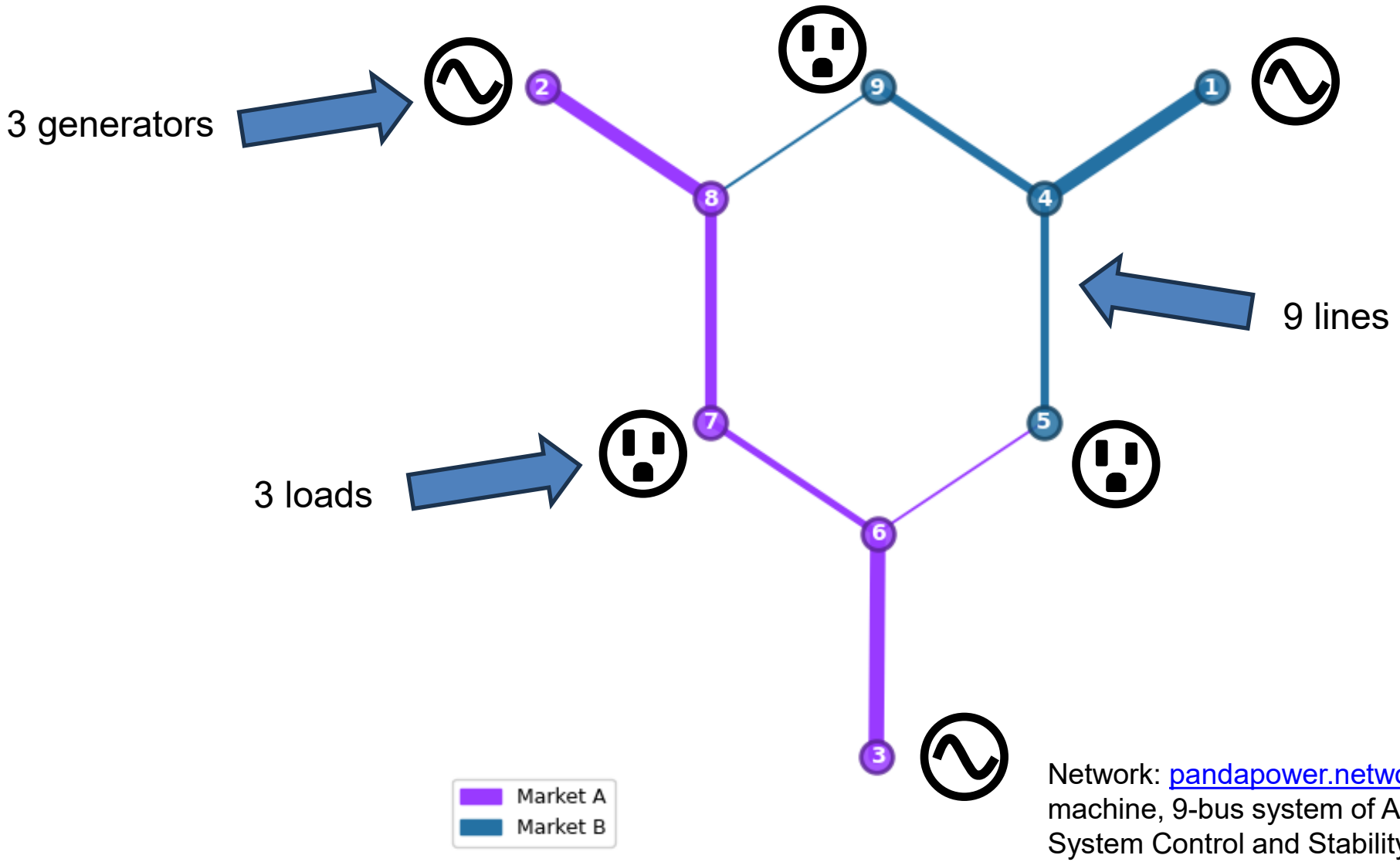
Paul Dockery
Market Strategy & Governance Manager

October 22, 2026

Western Loading Relief process example using an illustrative model

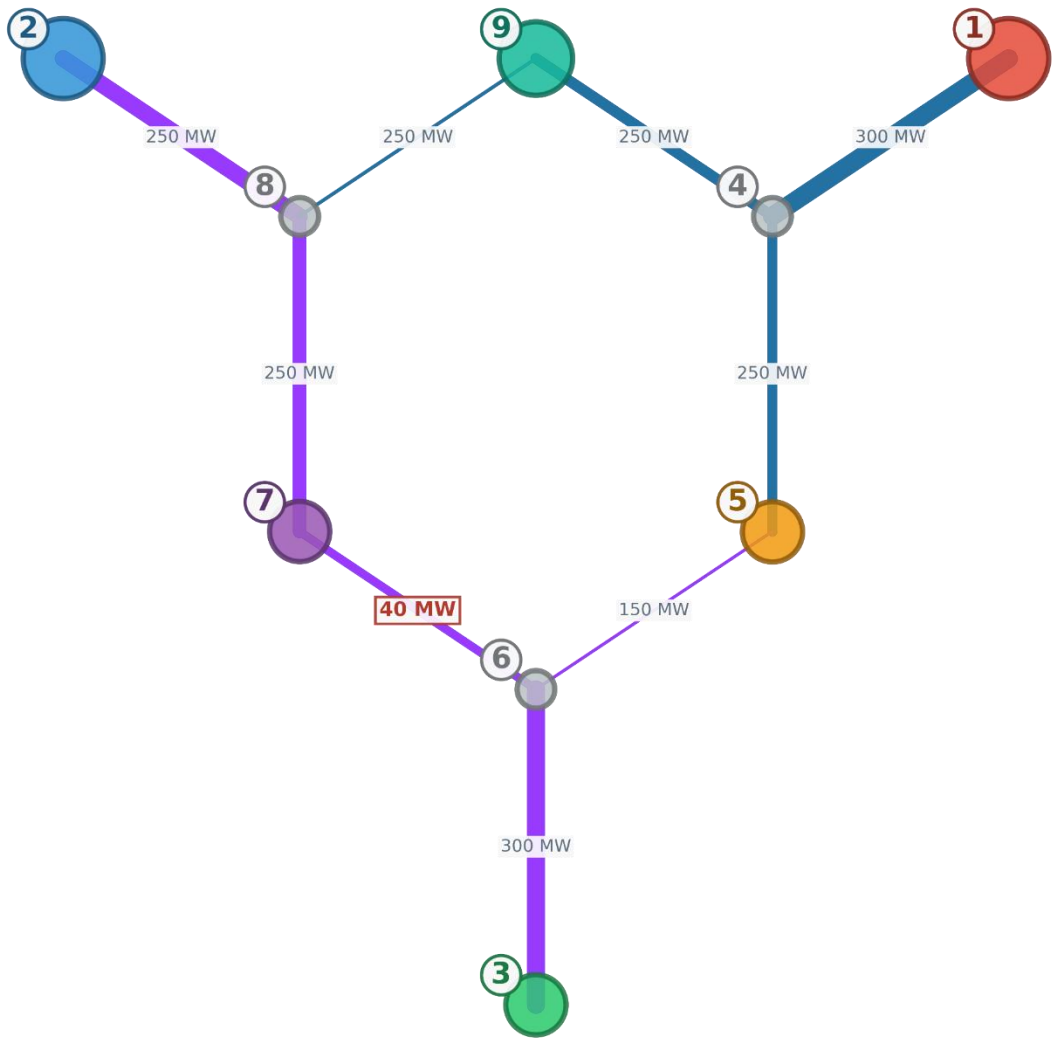
1. Example from August with two markets on a shared network that ignores neighbor's flow contribution on a binding element
2. Extension of the example where Western Loading Relief process is used to allocate relief obligation among tagged schedules
3. WLR process is used to allocate relief obligations for an example with both tagged schedules and gen-to-load
4. WLR process allocates relief obligations and markets are used to provide the relief

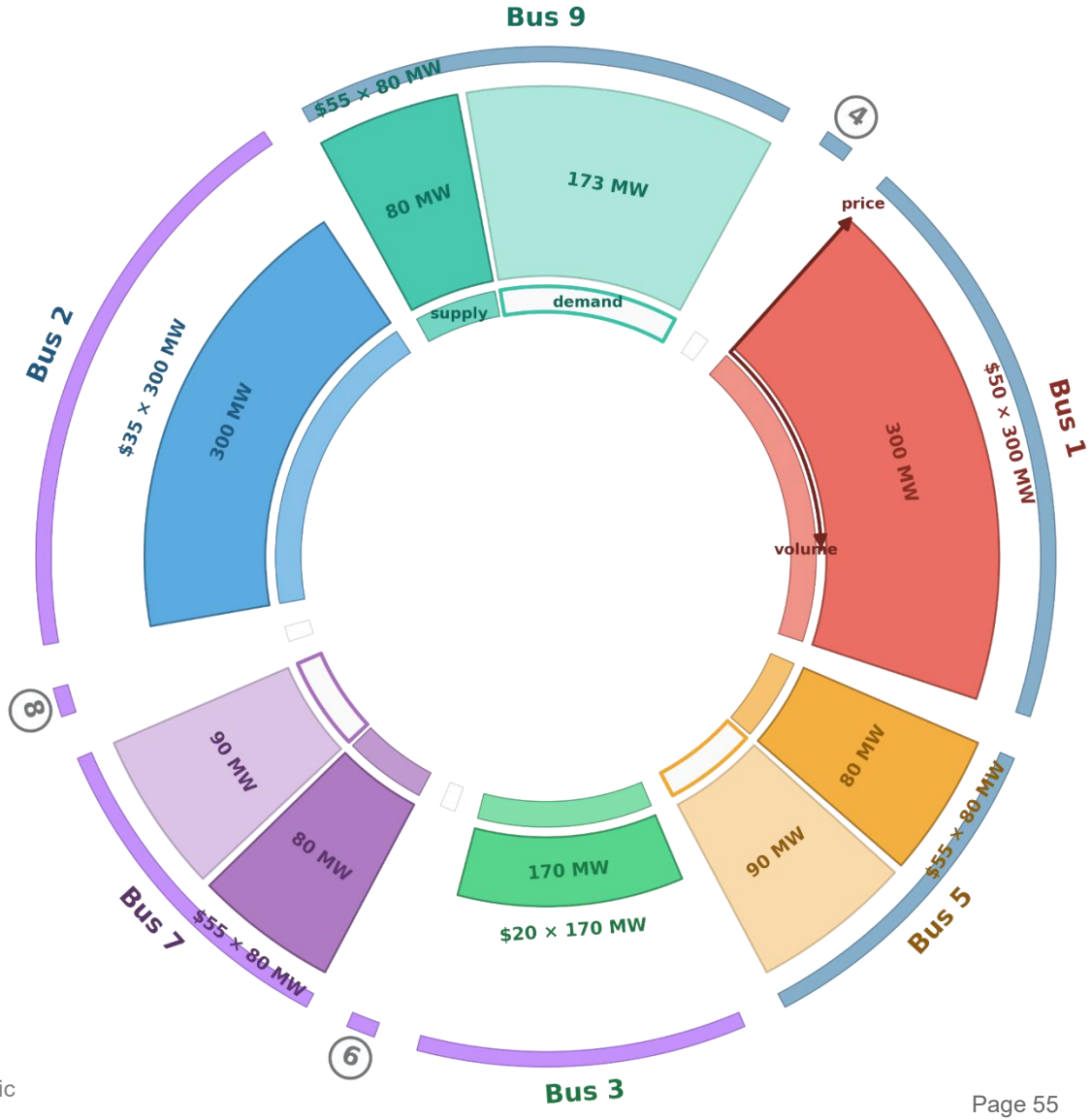
WSCC 9 Bus Test Case – Seams Example



Network: pandapower.networks.case9 after the WSCC 3-machine, 9-bus system of Anderson & Fouad, Power System Control and Stability (1977).

WSCC 9 Bus Test Case – Seams Example

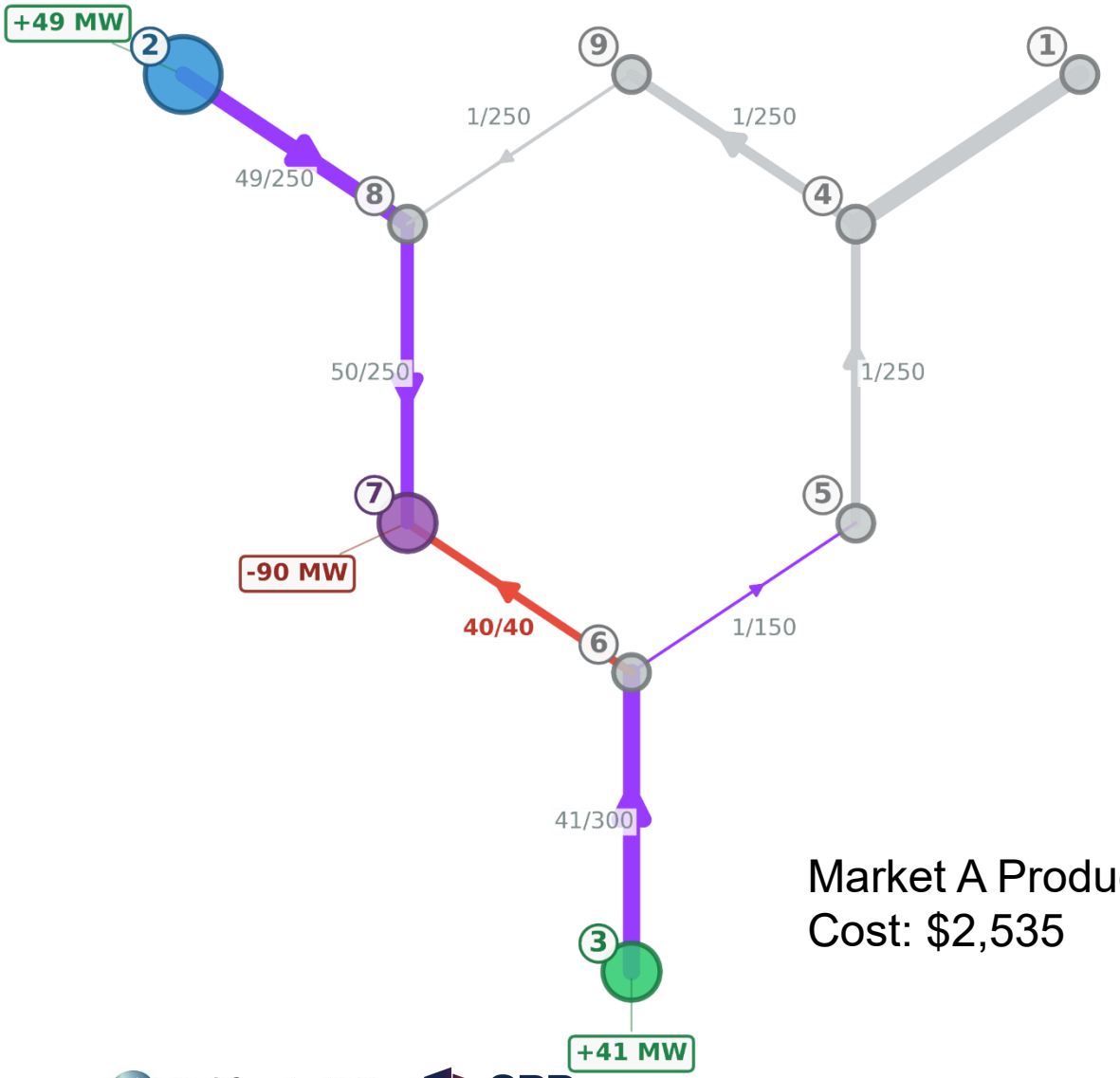




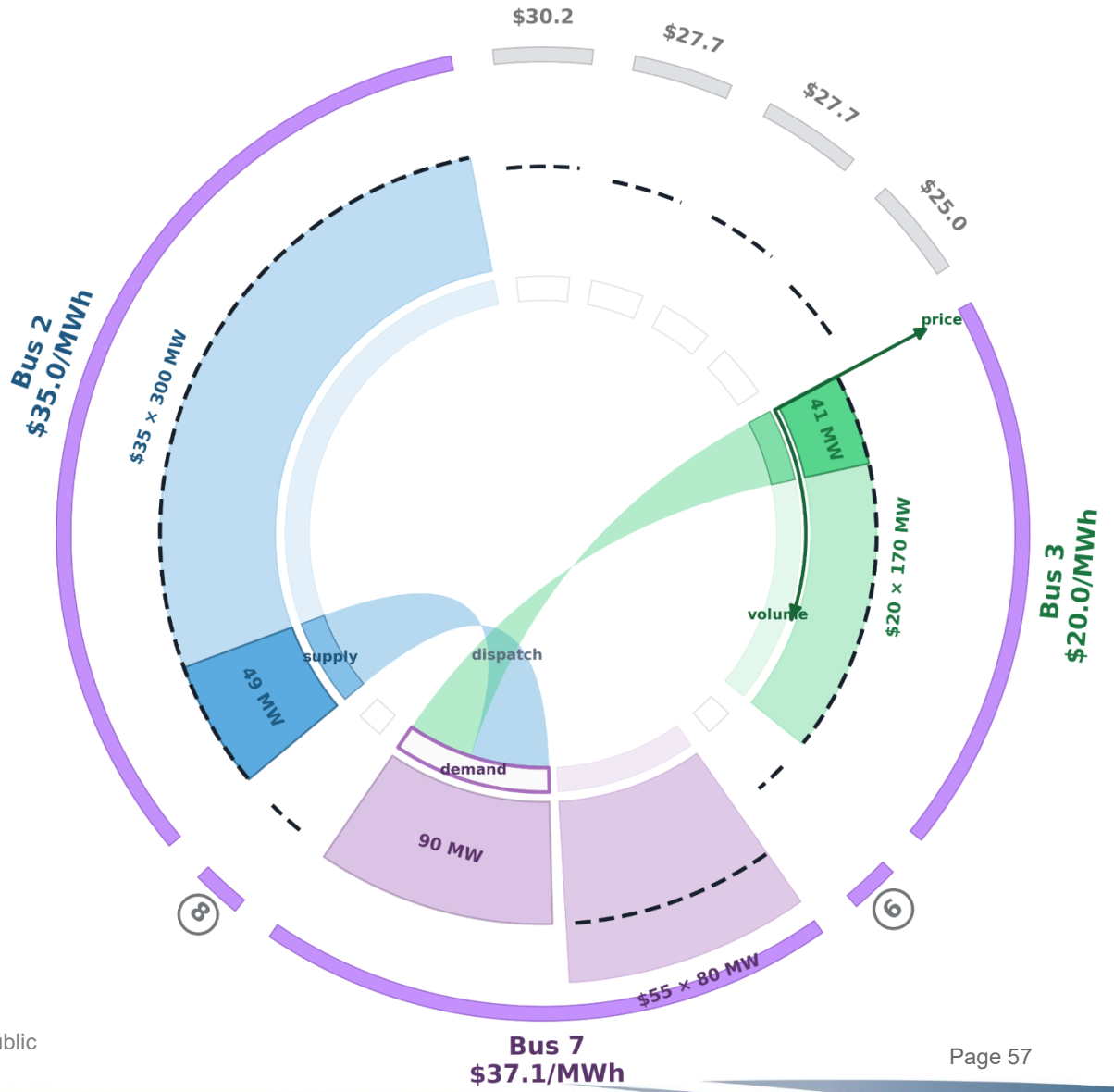
August Joint Workshop – Example #1 – no forecast of neighbor's flow contribution

- Each market clears independently without accounting for each other's use of the transmission network
- Market A has a binding constraint and finds the least cost dispatch of its system resources while respecting the limit
- Market B has no binding constraints and does not monitor binding constraint in Market A
- Power flow on the network based on combined dispatch overloads a line
- ***Market A redispatches to bring the line back into its limits***

Example #1 – no forecast of neighbor's flow contribution – Market A



Market A Production
Cost: $2,535



Example #1 – no forecast of neighbor's flow contribution – Market A

Marginal cost of serving load at bus 7

+ Bus 2 Gen

↑ 1.14 MW @ $35/MWh

= $39.90

- Bus 3 Gen

↓ 0.14 MW @ $20/MWh

= $2.80

△ Bus 7: $39.90 - $2.80 = $37.10



Example #1 – no forecast of neighbor's flow contribution – Market A

Marginal cost of flowing on the line from 6 to 7

+ Bus 2 Gen

↑ 1.34 MW @ $35/MWh

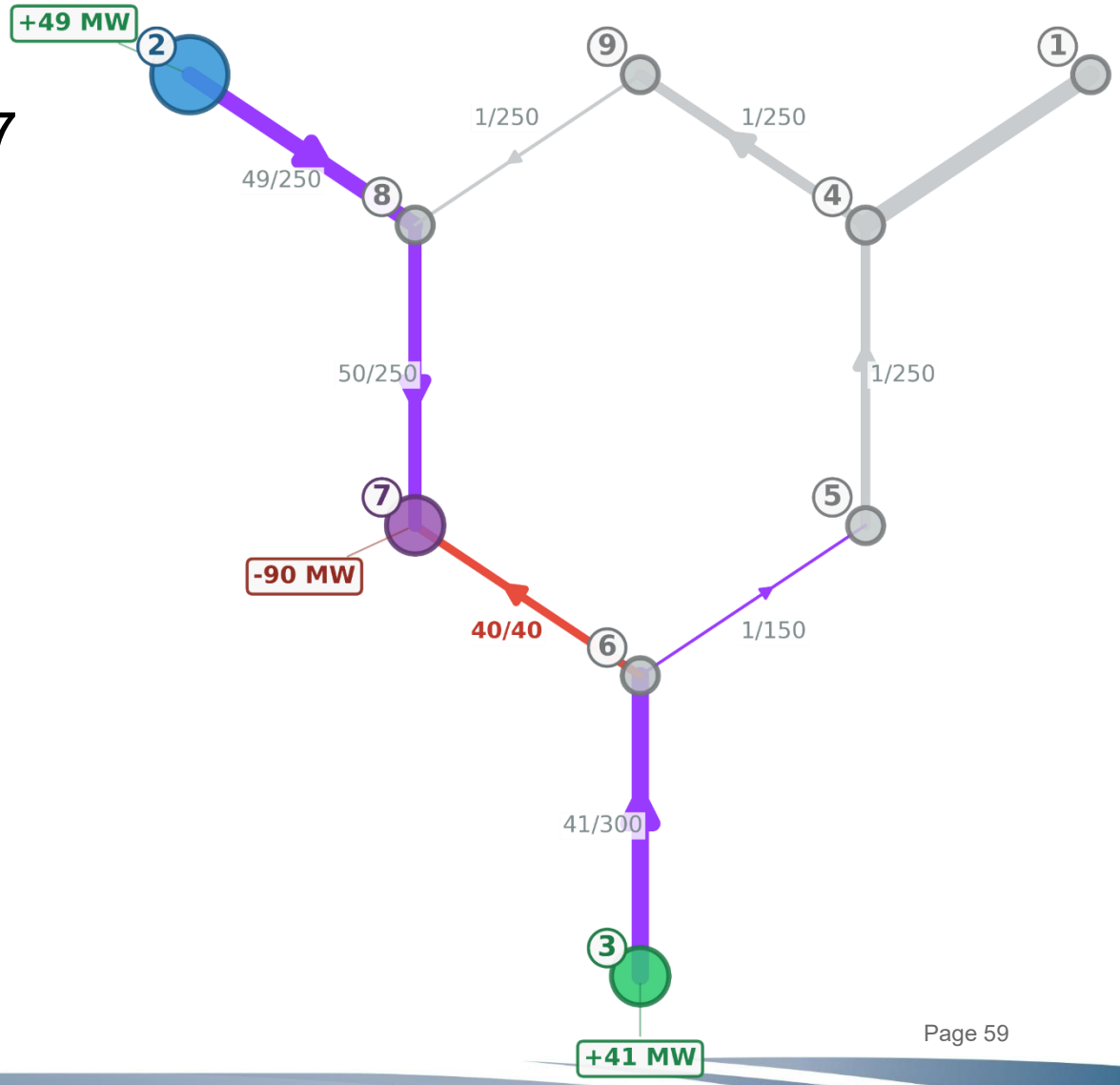
= $46.90

- Bus 3 Gen

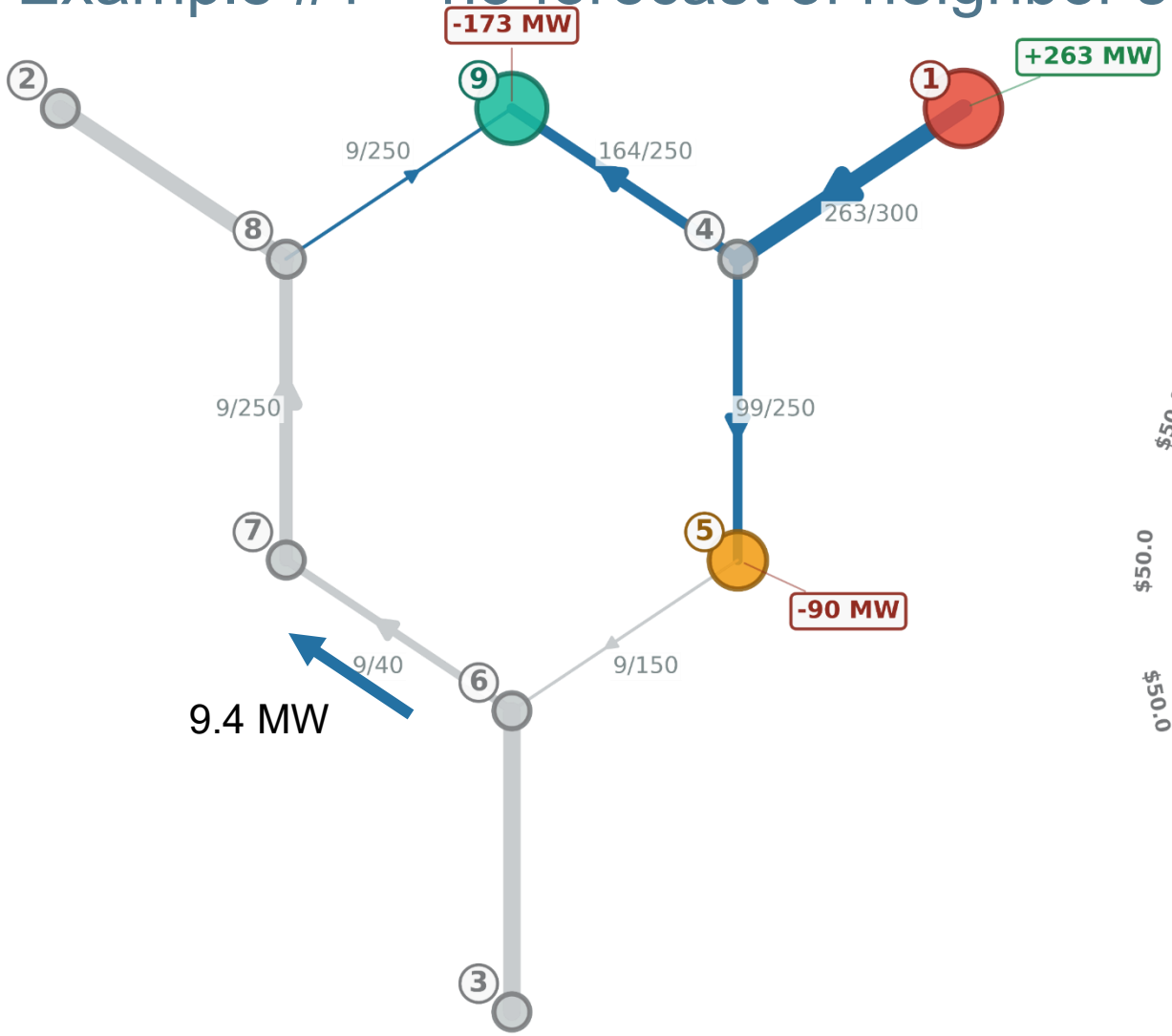
↓ 1.34 MW @ $20/MWh

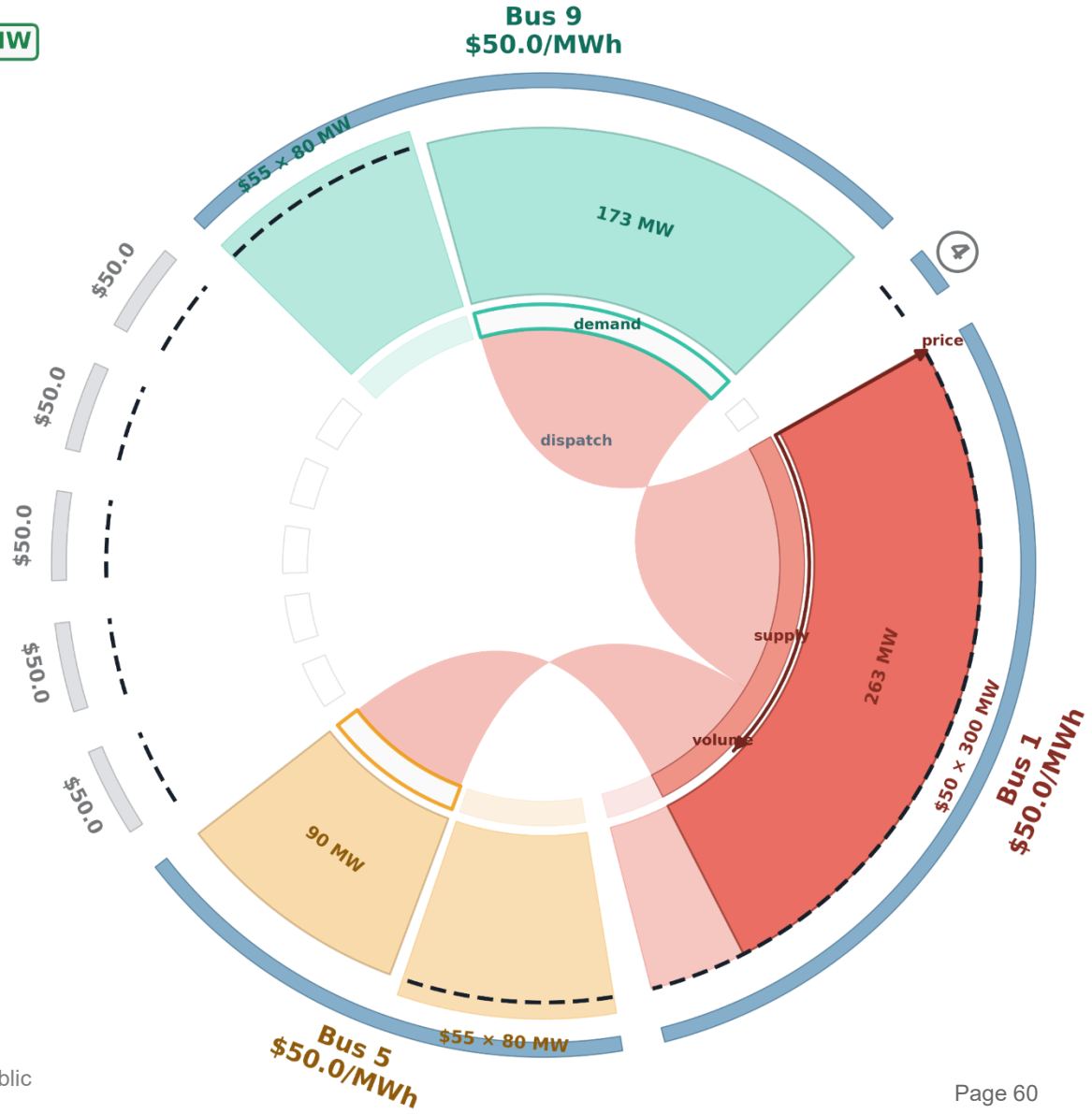
= $26.80

△ Line 6-7: $46.90 - $26.80 = $20.10

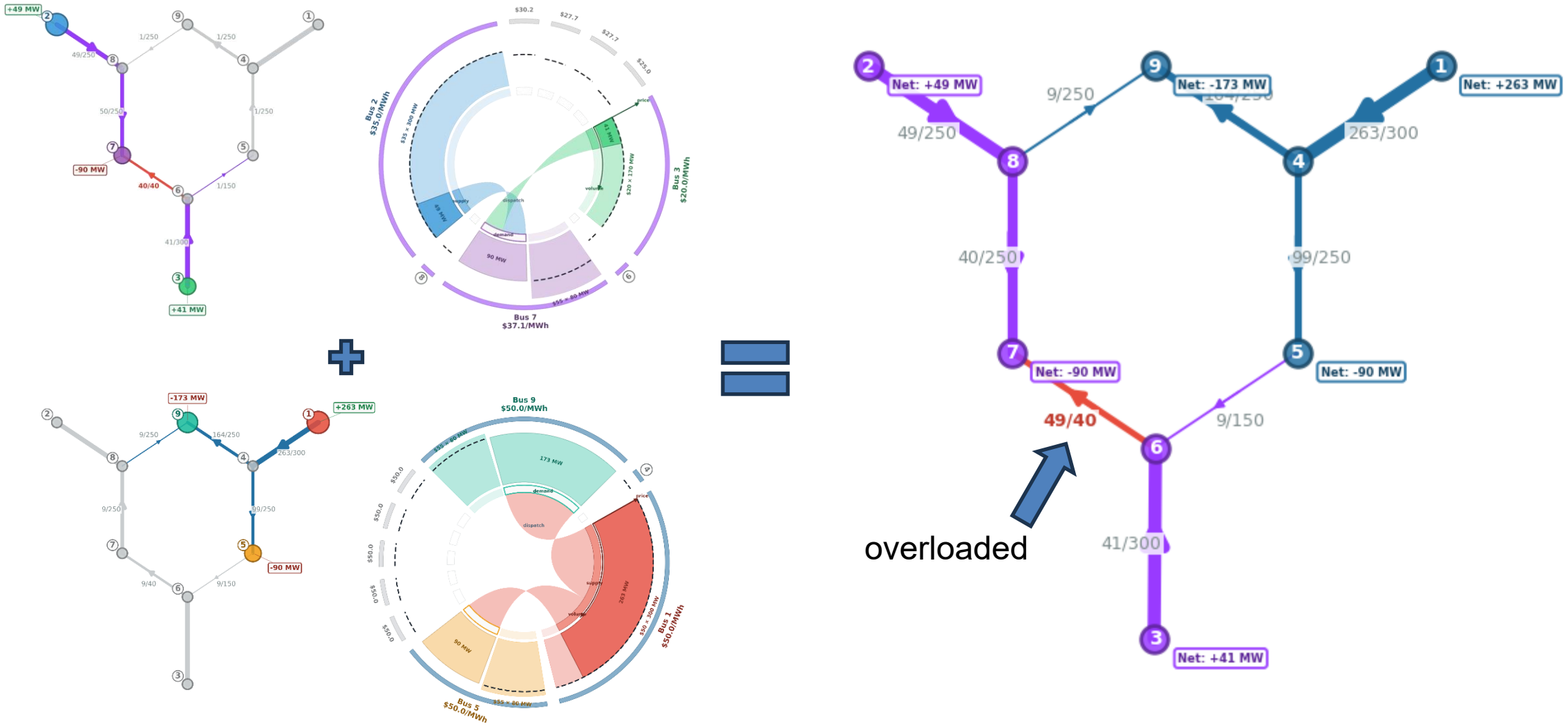


Example #1 – no forecast of neighbor's flow contribution – Market B





Example #1 – no forecast of neighbor's flow contribution - combined



Example #1 – no forecast of neighbor's flow contribution – redispatch

Market A Production Cost: $2,725

Independent run: $2,535

+ Bus 2 Gen

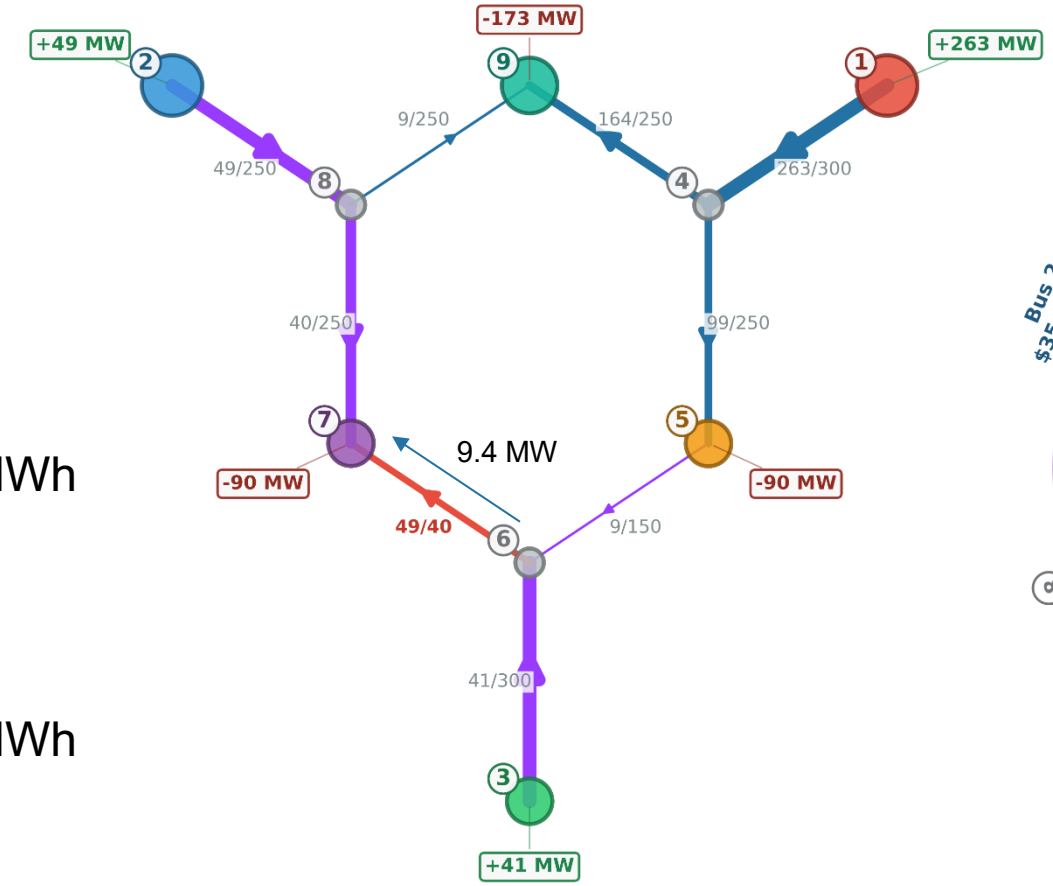
↑ 12.66 MW @ $35/MWh

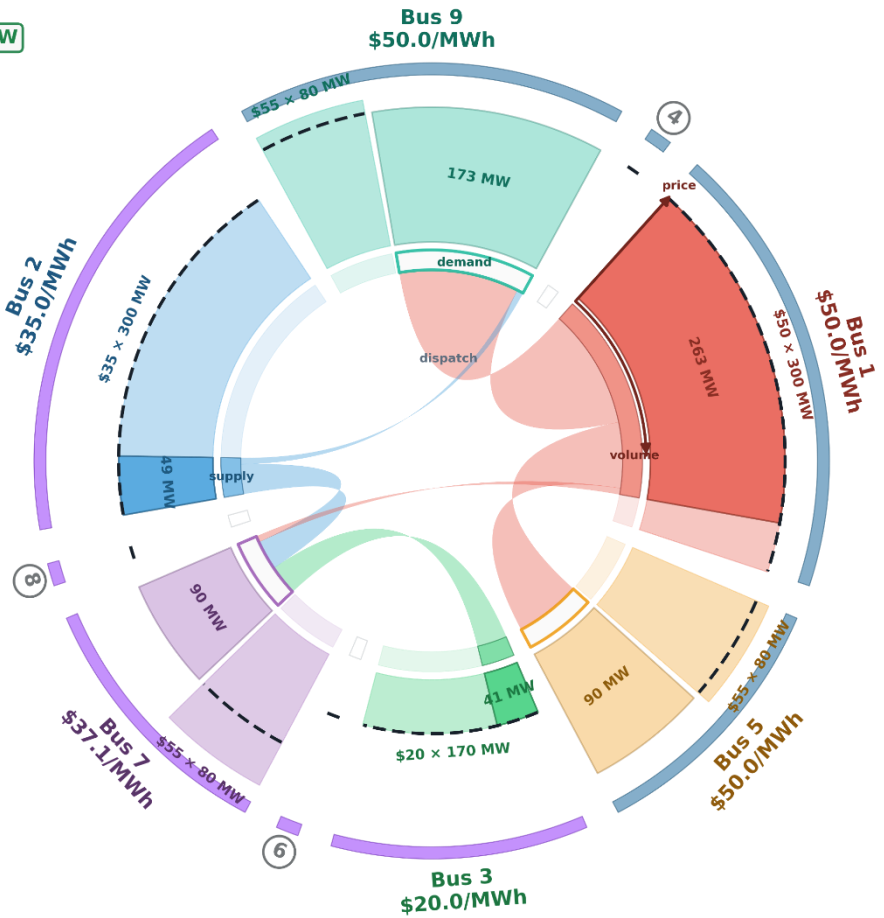
= $443

- Bus 3 Gen

↓ 12.66 MW @ $20/MWh

= $253



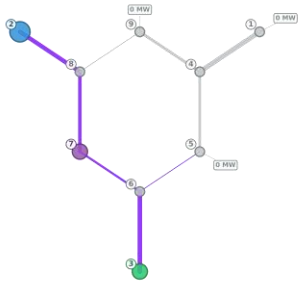


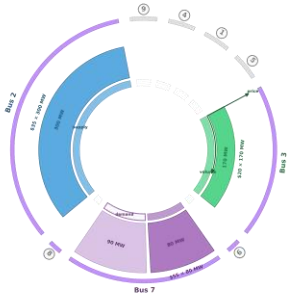
△ Market A production cost: $443 - $253 = $190

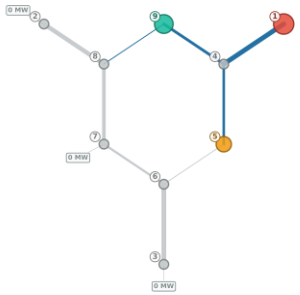
= 9.4 MW @ $20.10/MWh = $190

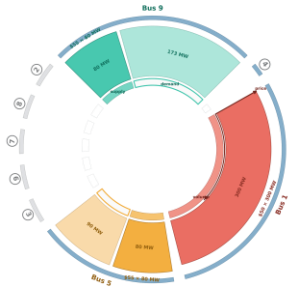
Parallel process considerations

Inputs

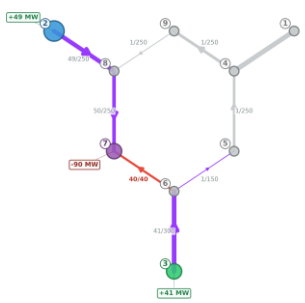


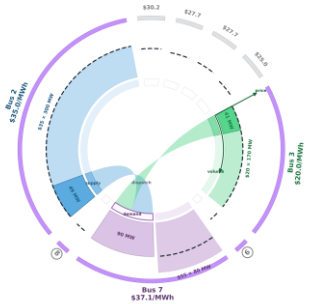


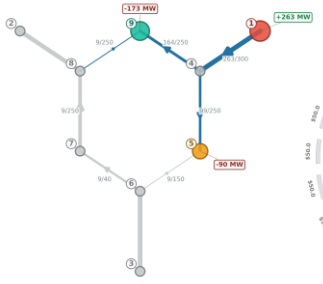


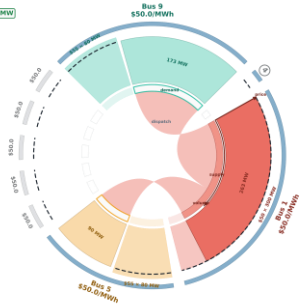


Optimization

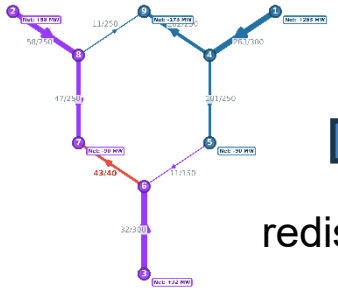




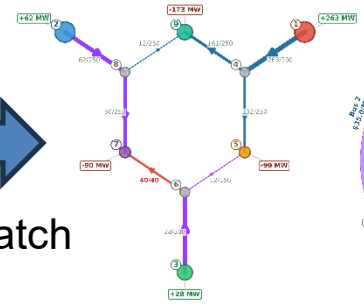


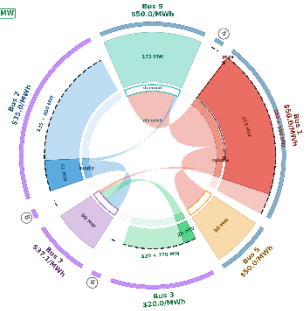


Reconciliation



redispatch

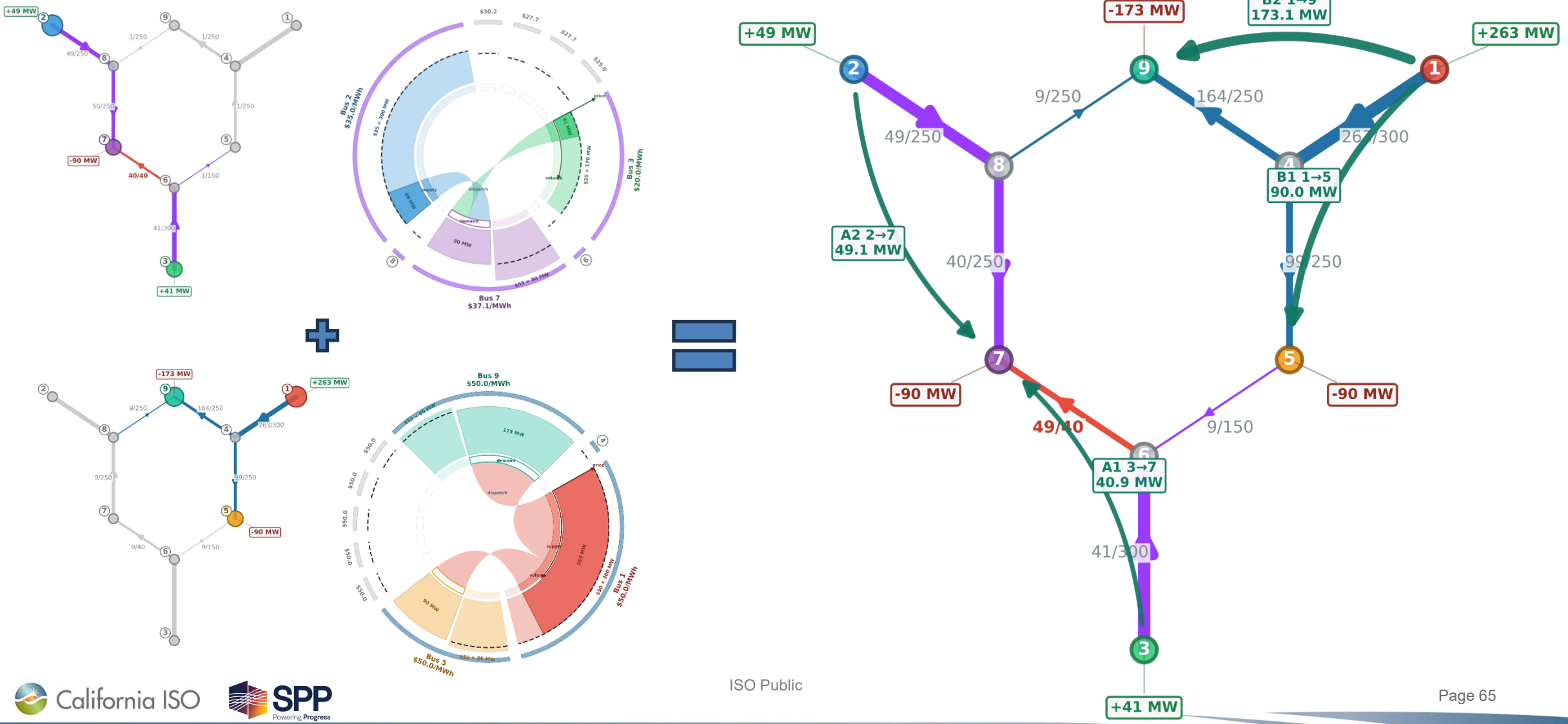




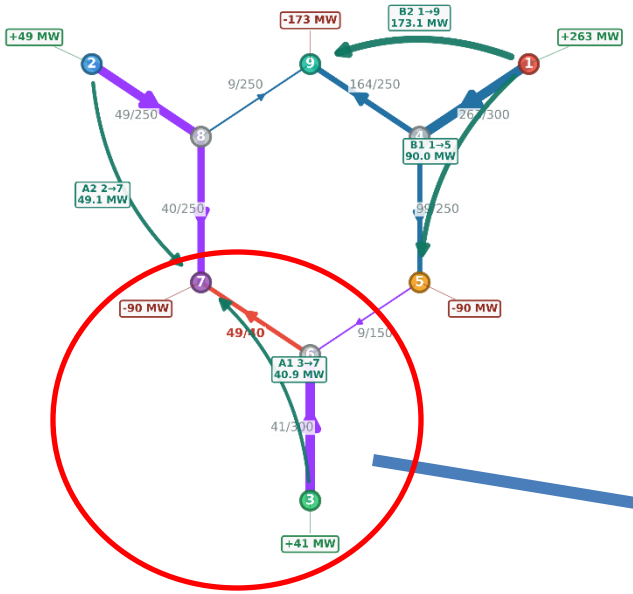
Example #2 – Reconciliation with WLR process – everything tagged

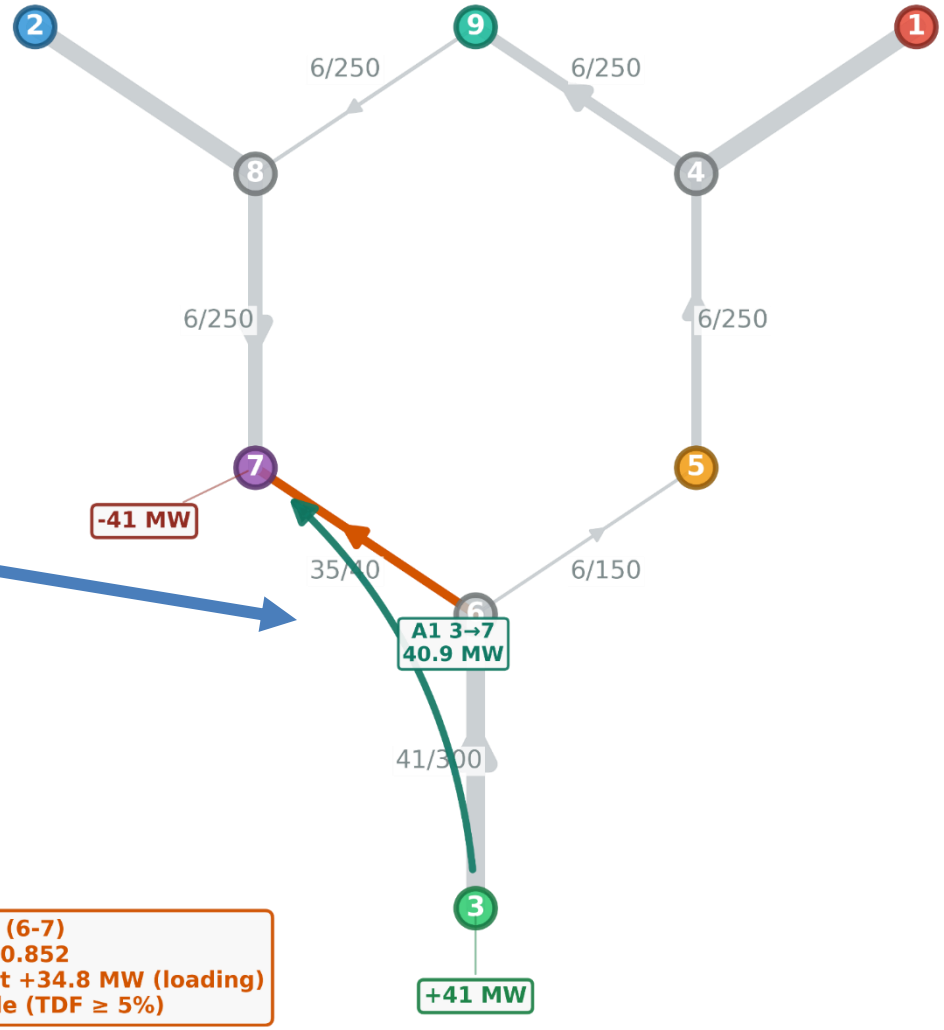
- Start with Example #1 market dispatch
- Illustrative tagged schedules based on example #1 market dispatch
- Power flow on the network based on combined dispatch overloads a line
- ***WLR process assigns a relief obligation*** to bring the line under its limits by reducing tagged schedules

Example #1 – Reconcile dispatch with WLR process



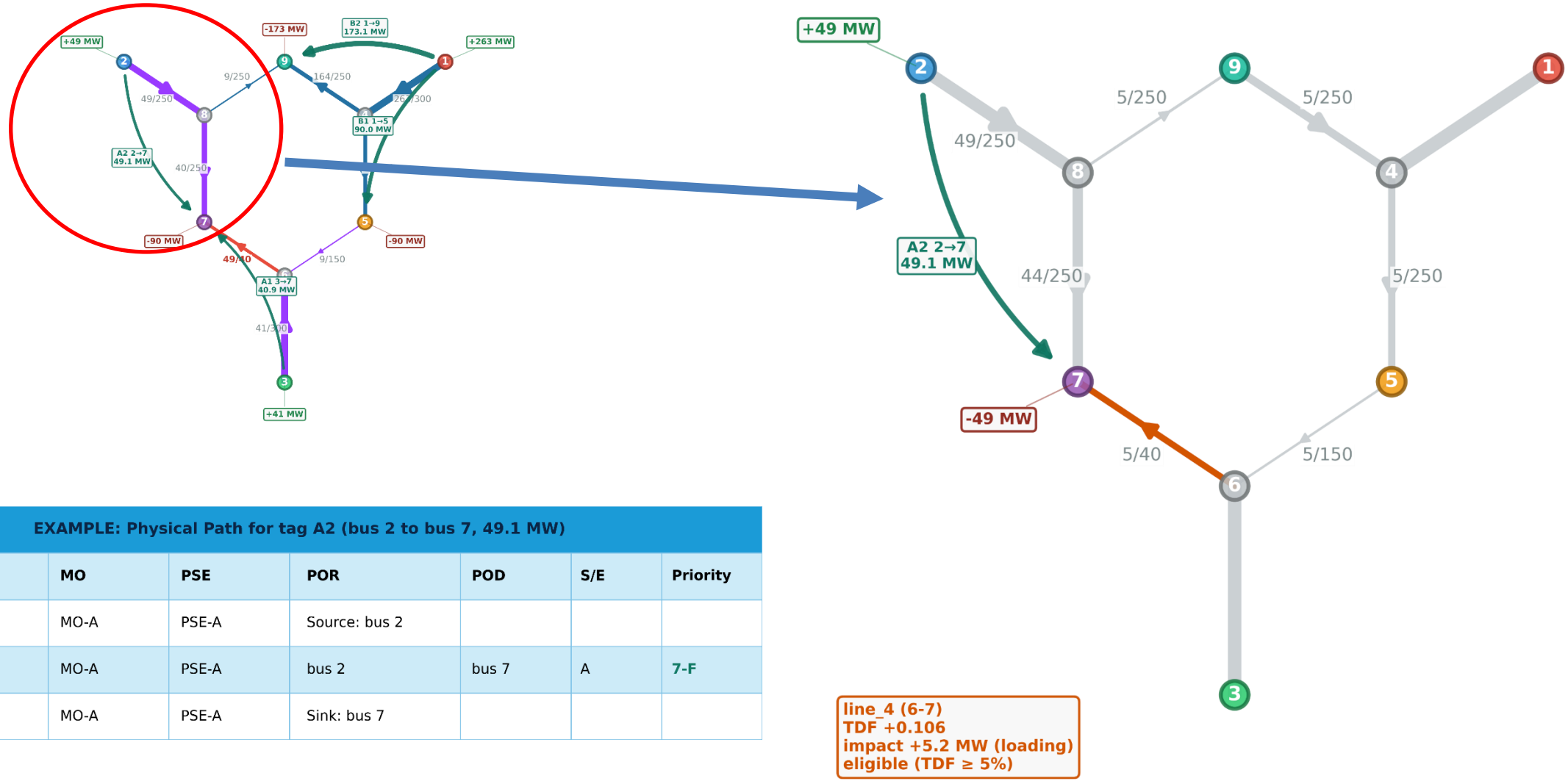
Example #1 – Enhanced Curtailment Calculator – Tag 3->7





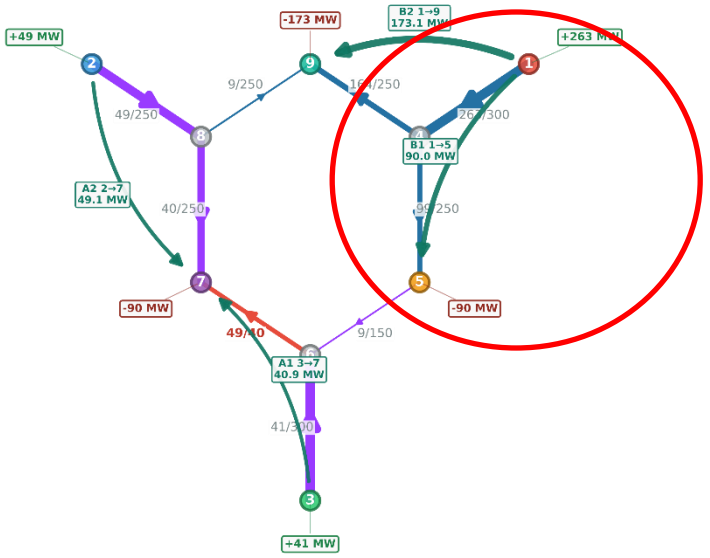
EXAMPLE: Physical Path for tag A1 (bus 3 to bus 7, 40.9 MW)							
BA	**TSP**	**MO**	**PSE**	**POR**	**POD**	**S/E**	**Priority**
A		MO-A	PSE-A	Source: bus 3			
	A	MO-A	PSE-A	bus 3	bus 7	A	**7-F**
A		MO-A	PSE-A	Sink: bus 7			

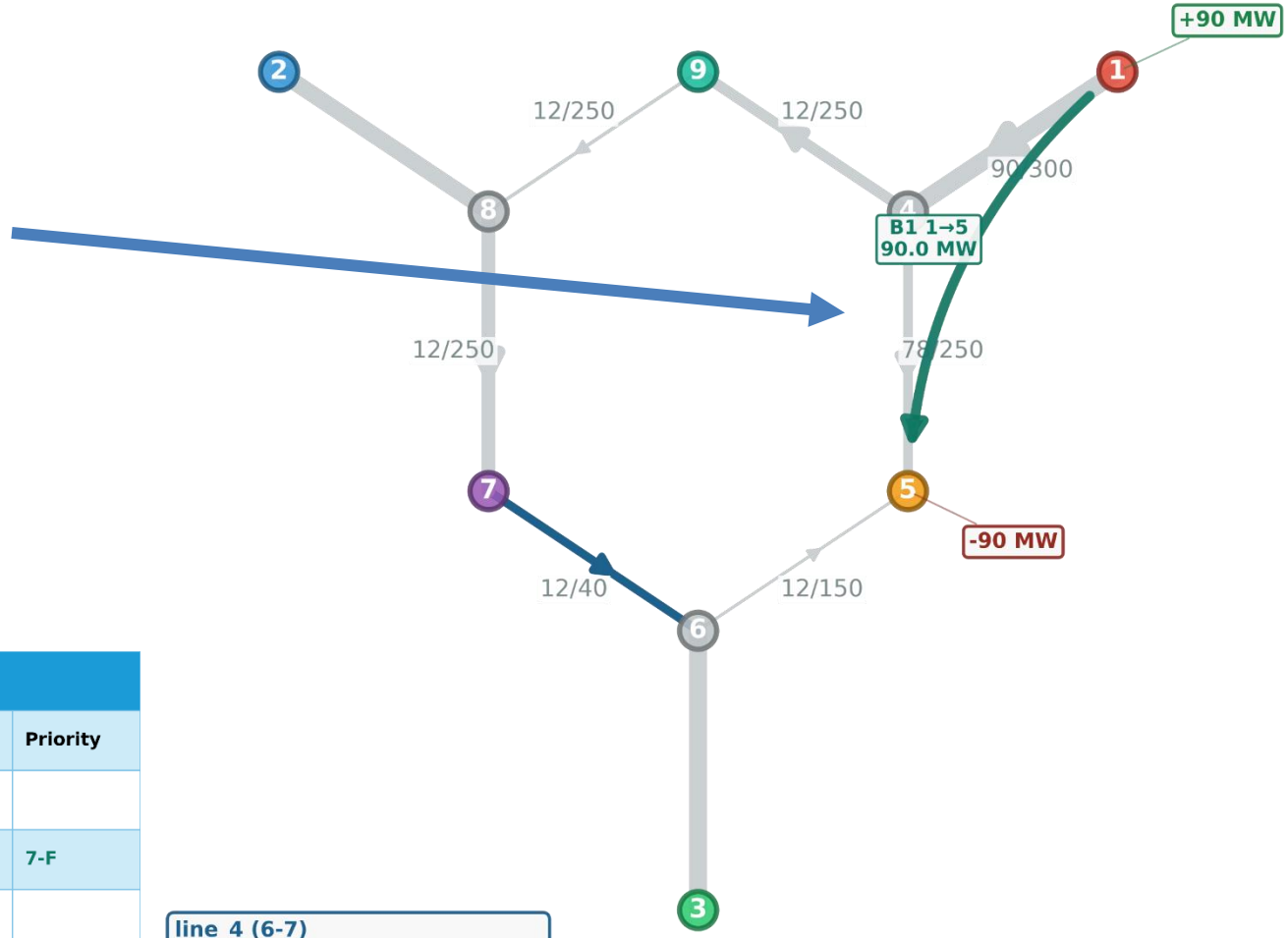
Example #1 – Enhanced Curtailment Calculator – Tag 2->7



EXAMPLE: Physical Path for tag A2 (bus 2 to bus 7, 49.1 MW)							
BA	**TSP**	**MO**	**PSE**	**POR**	**POD**	**S/E**	**Priority**
A		MO-A	PSE-A	Source: bus 2			
	A	MO-A	PSE-A	bus 2	bus 7	A	**7-F**
A		MO-A	PSE-A	Sink: bus 7			

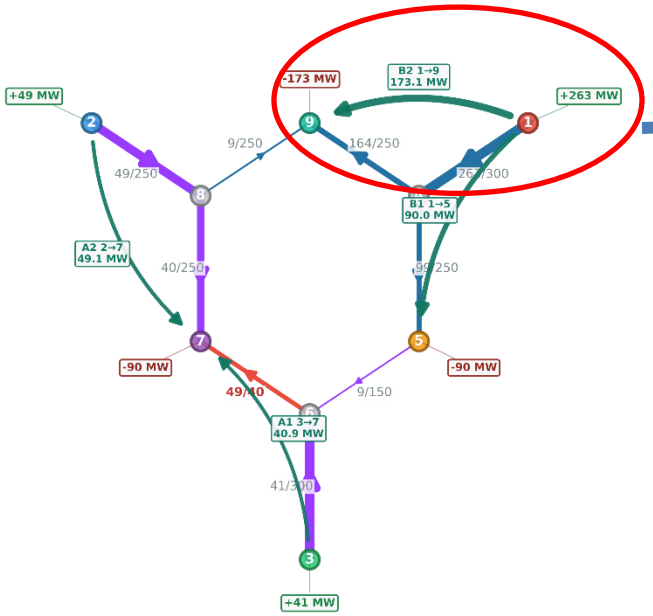
Example #1 – Enhanced Curtailment Calculator – Tag 1->5

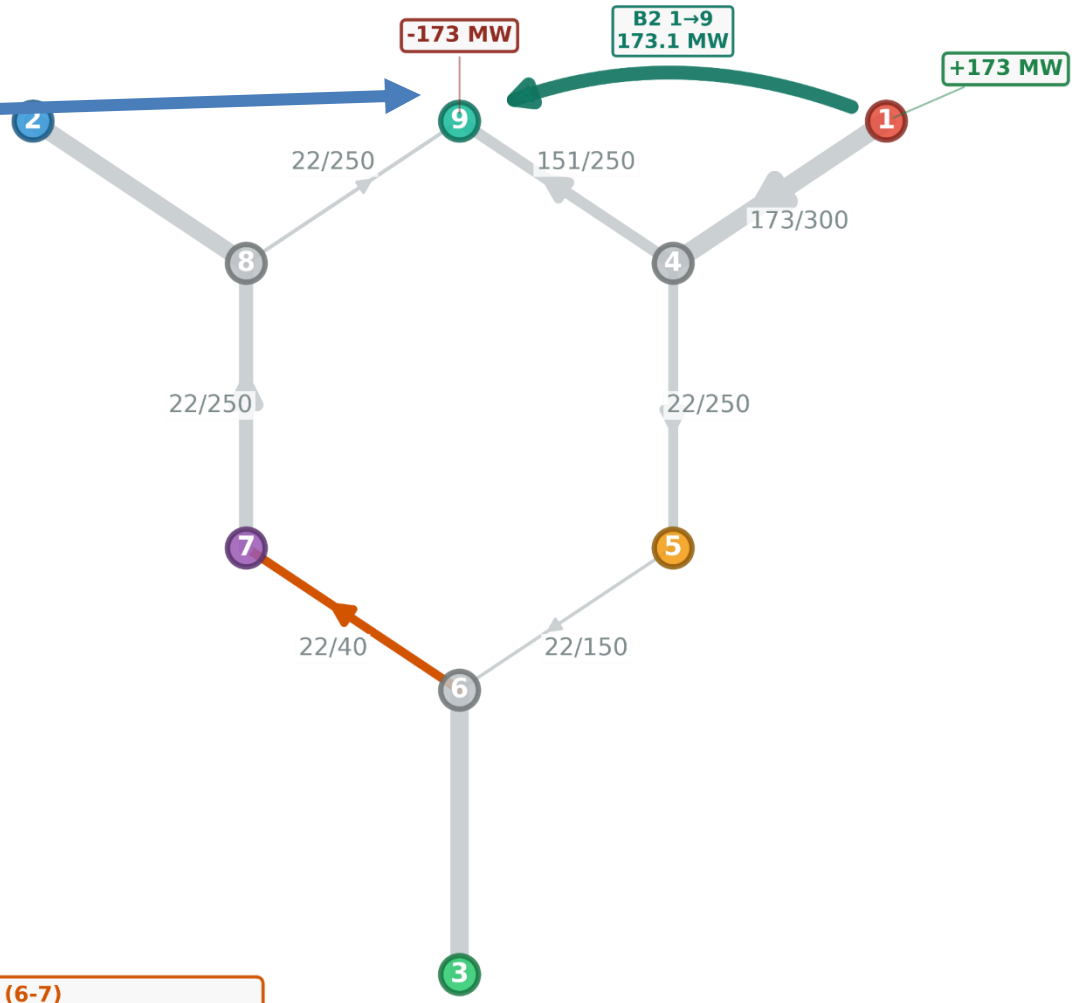




EXAMPLE: Physical Path for tag B1 (bus 1 to bus 5, 90.0 MW)							
BA	**TSP**	**MO**	**PSE**	**POR**	**POD**	**S/E**	**Priority**
B		MO-B	PSE-B	Source: bus 1			
	B	MO-B	PSE-B	bus 1	bus 5	B	**7-F**
B		MO-B	PSE-B	Sink: bus 5			

Example #1 – Enhanced Curtailment Calculator – Tag 1->9

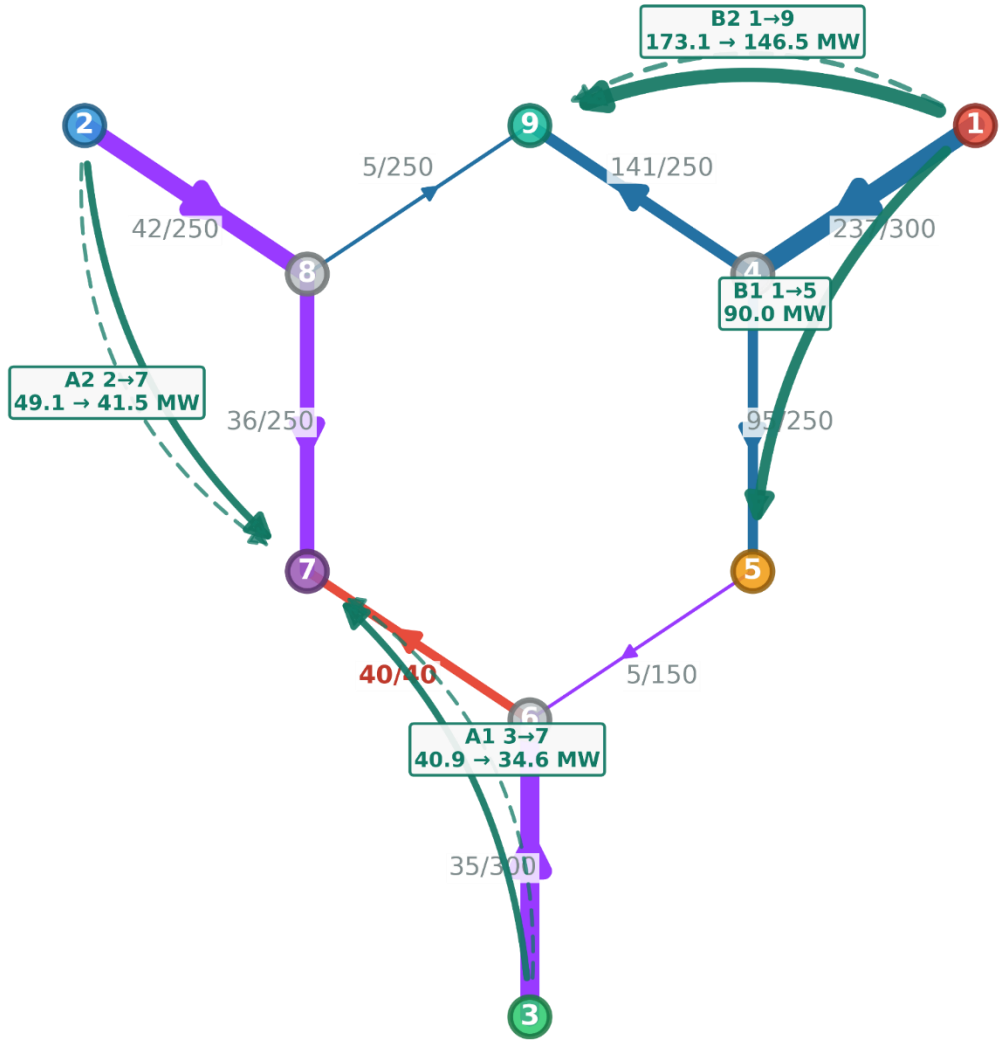




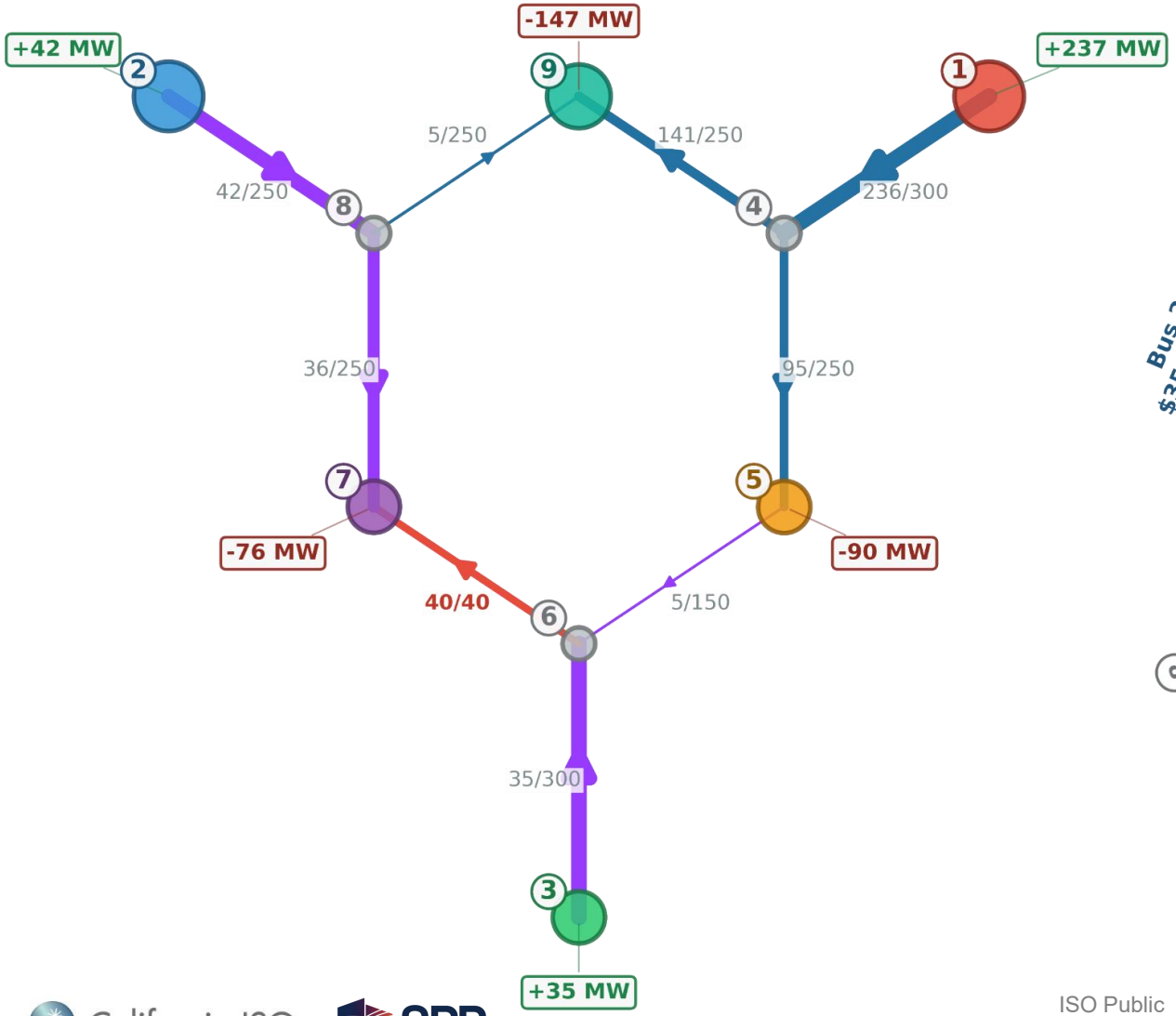
EXAMPLE: Physical Path for tag B2 (bus 1 to bus 9, 173.1 MW)							
BA	**TSP**	**MO**	**PSE**	**POR**	**POD**	**S/E**	**Priority**
B		MO-B	PSE-B	Source: bus 1			
	B	MO-B	PSE-B	bus 1	bus 9	B	**7-F**
B		MO-B	PSE-B	Sink: bus 9			

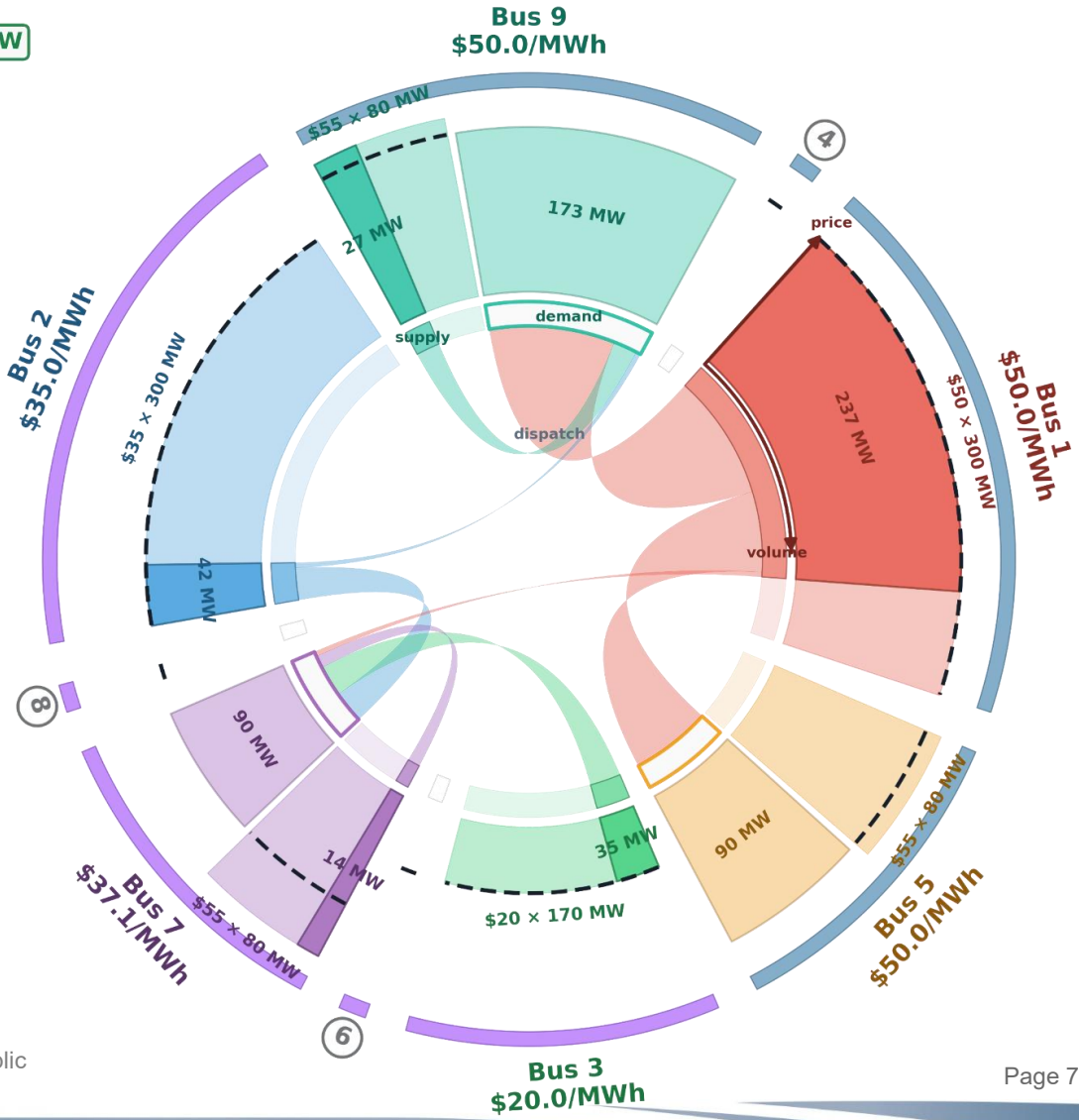
Calculation of relief

Tag	Priority	Scheduled (MW)	Impact (MW)	Share of relief	Relief (MW)	Curtailment or redispatch (MW)	Curtailed to (MW)
A1 (3→7)	7-F	40.9	34.8	56.5%	5.34	6.3	34.6
A2 (2→7)	7-F	49.1	5.2	8.4%	0.80	7.5	41.5
B2 (1→9)	7-F	173.1	21.6	35.1%	3.31	26.5	146.5
B1 (1→5, counterflow)	7-F	90.0	-12.2	not eligible	0.00	0.0	90.0
Total		**353.0**	**49.4**	**100.0%**	**9.45**	**40.3**	**312.7**



Resulting dispatch

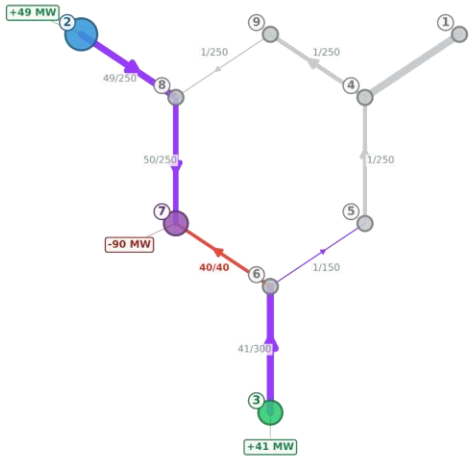


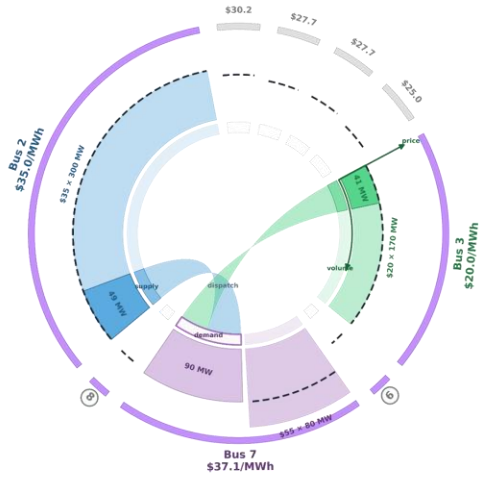


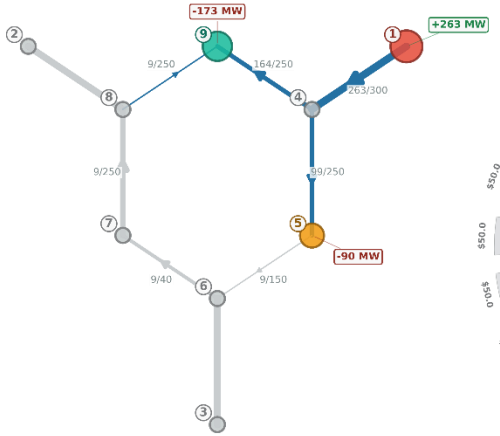
Example #3 – Reconciliation with WLR process – Gen-to-Load example

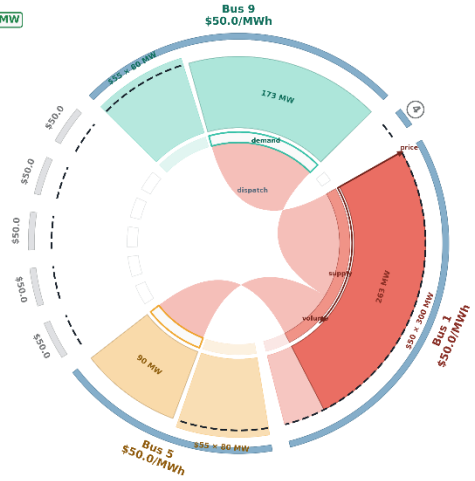
- Start with Example #1 market dispatch
- Illustrative combination of tagged schedules and untagged gen-to-load based on example #1 market dispatch
- Power flow on the network based on combined dispatch overloads a line
- ***WLR process assigns a relief obligation*** to bring the line under its limits by reducing tagged schedules and gen-to-load

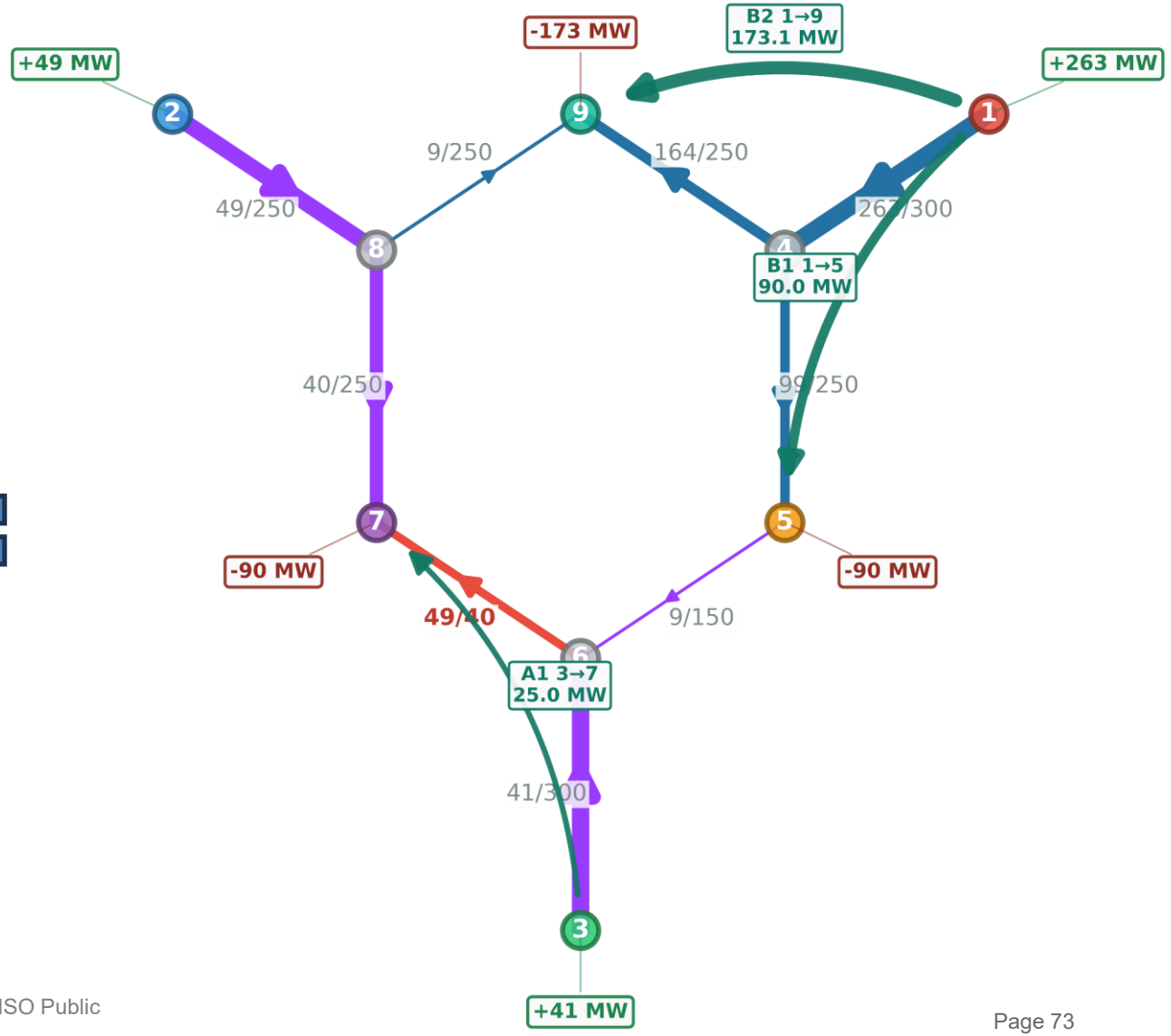
Example #3 – Reconcile dispatch with WLR process



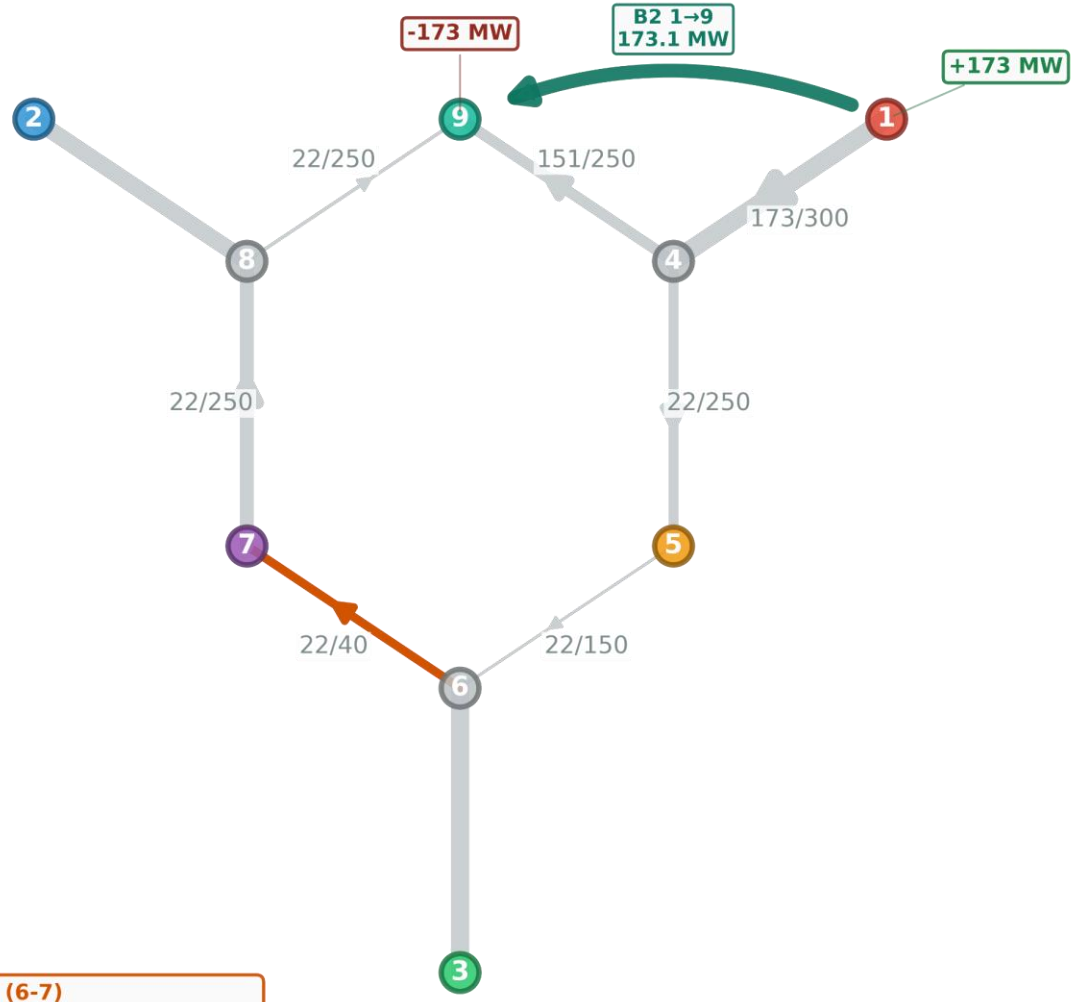


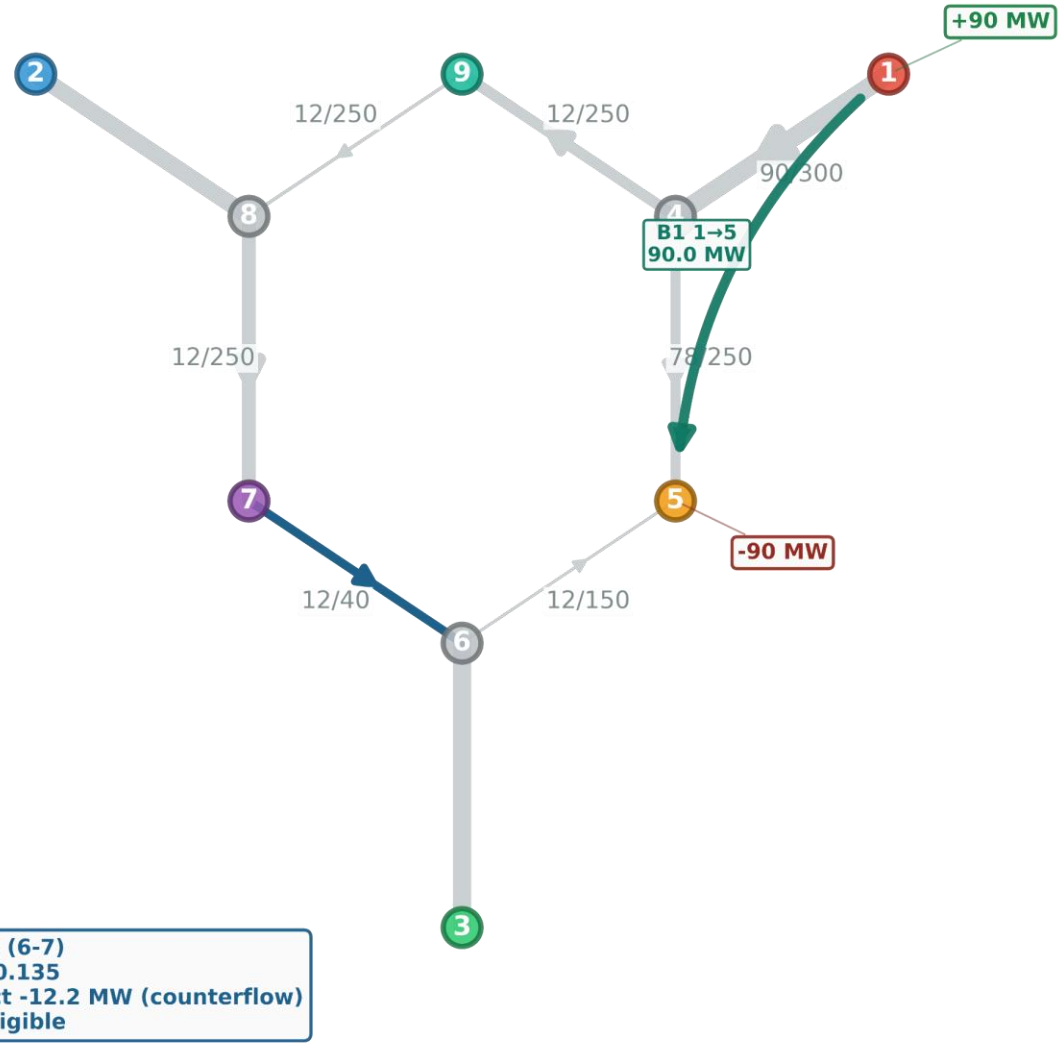




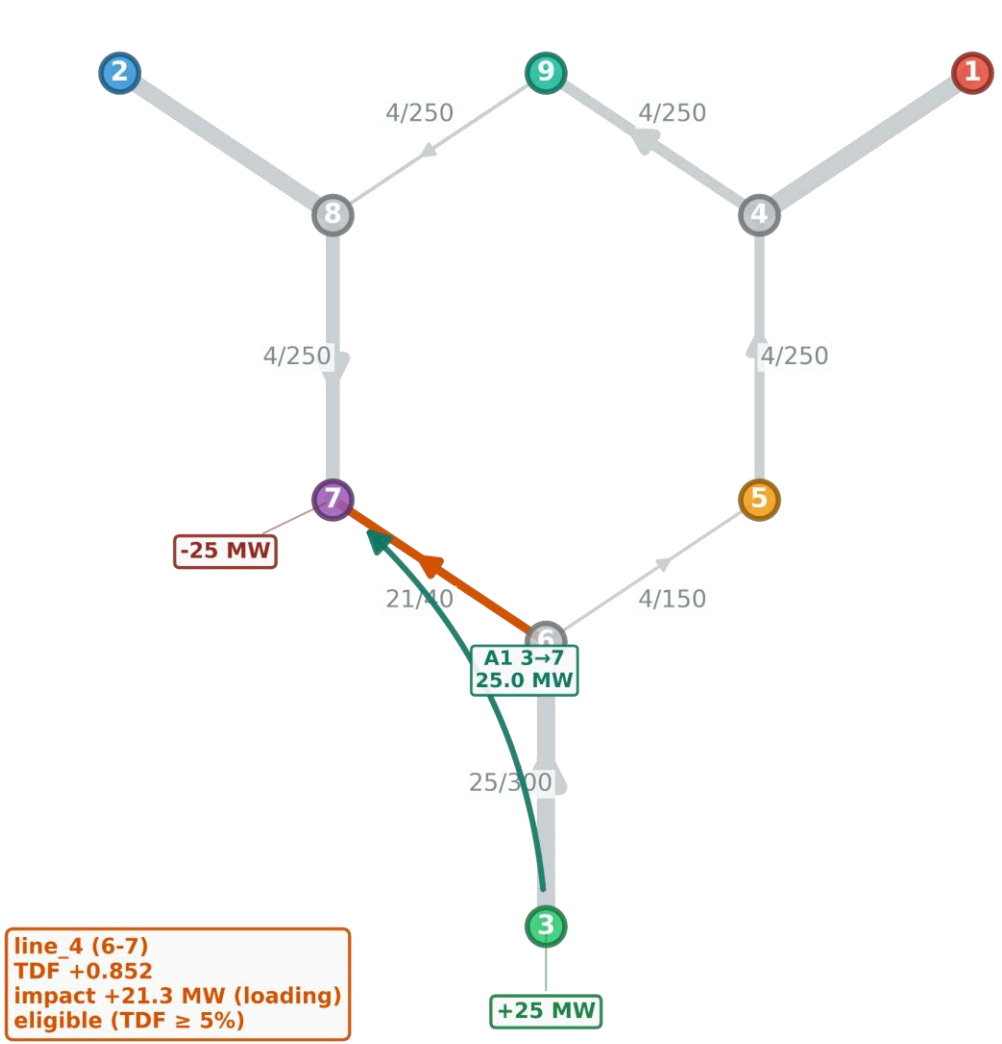


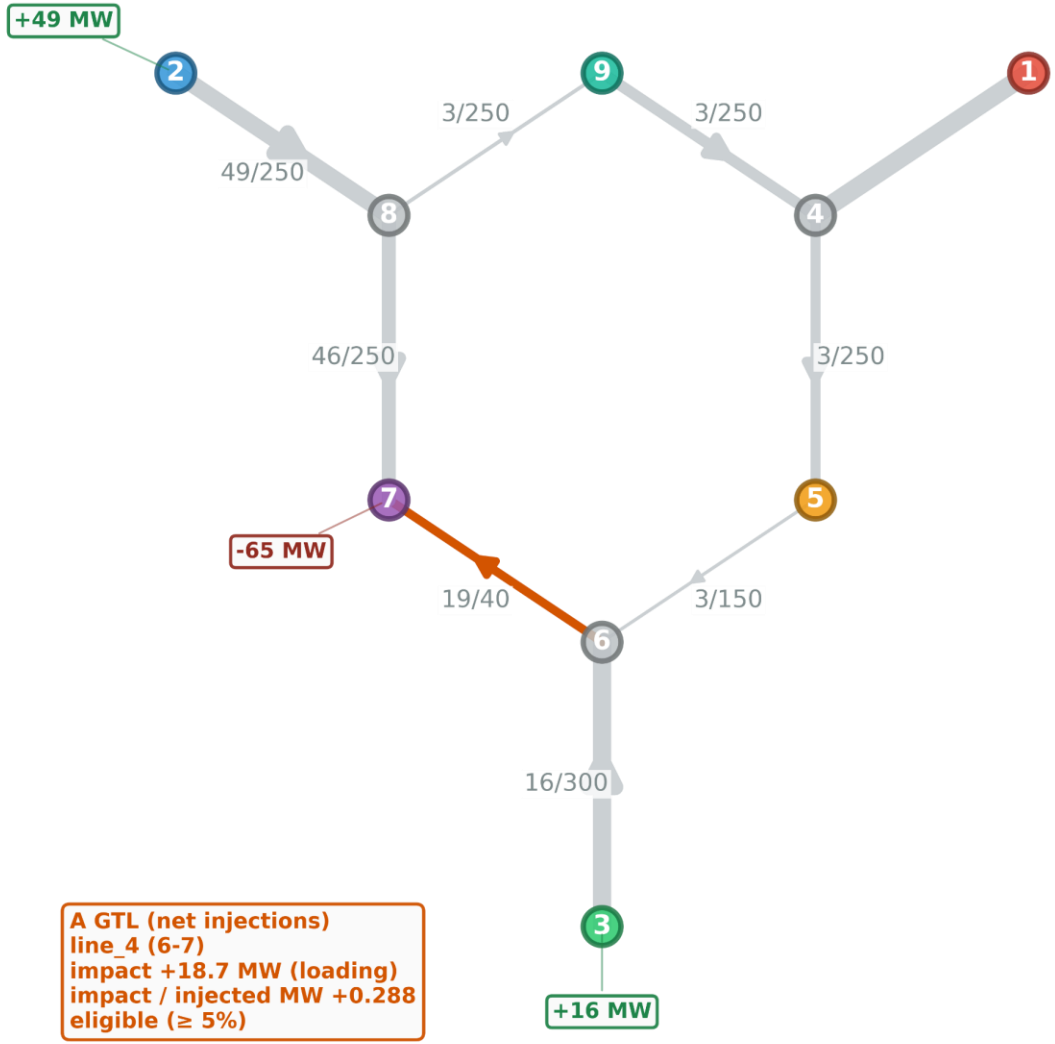
Example #3 – Market B tagged schedules





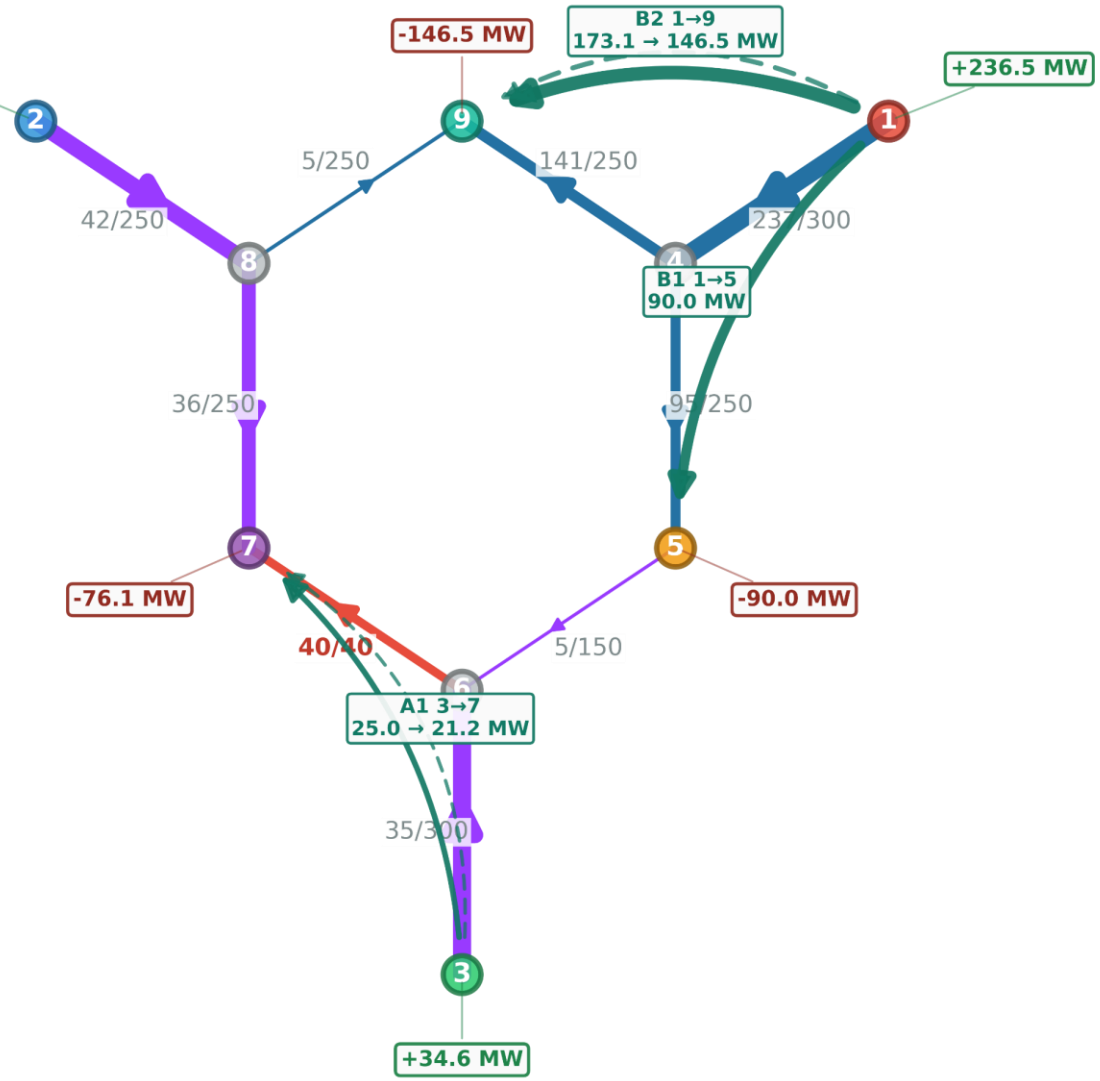
Example #3 – Market A tagged schedules & gen-to-load



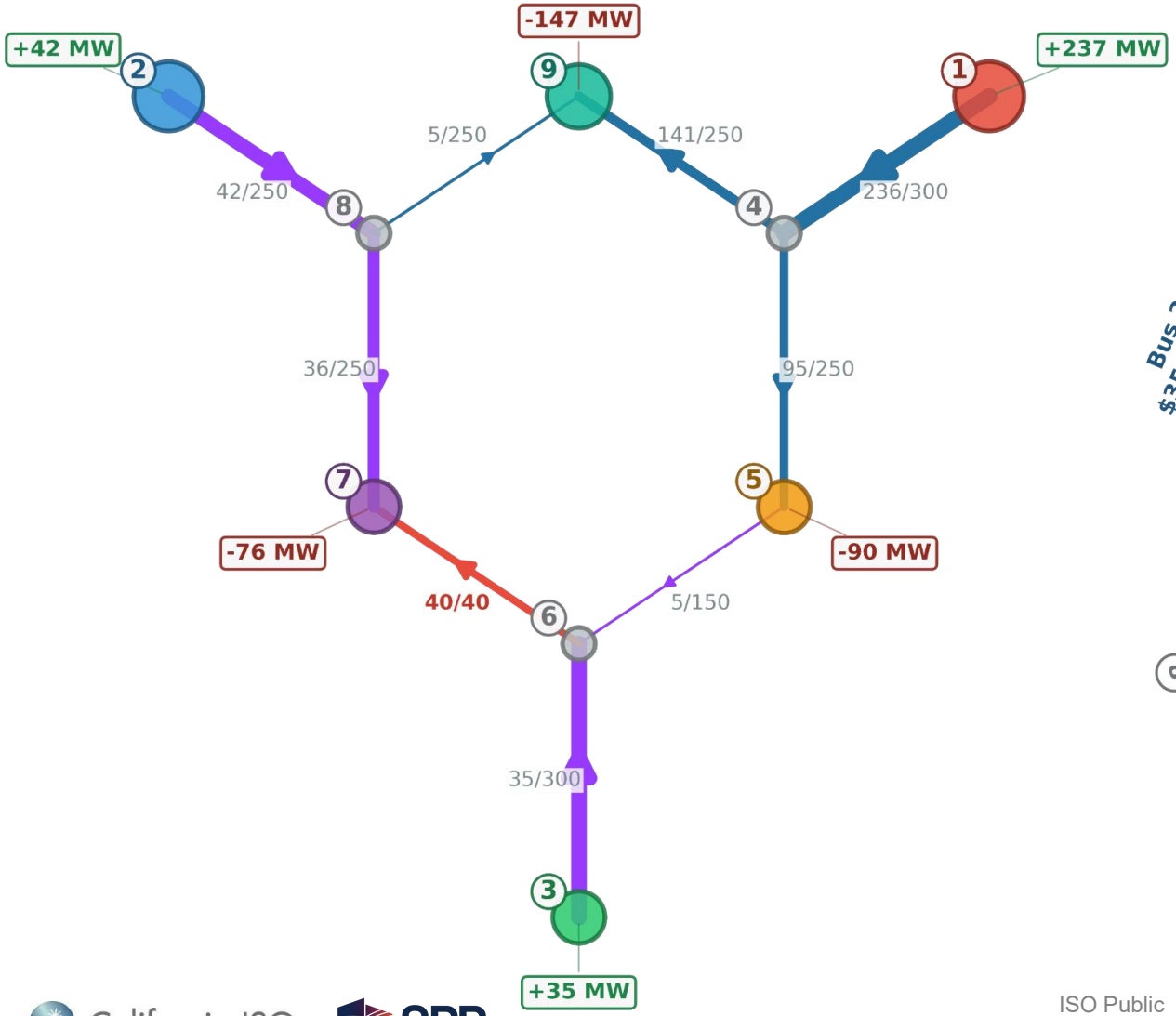


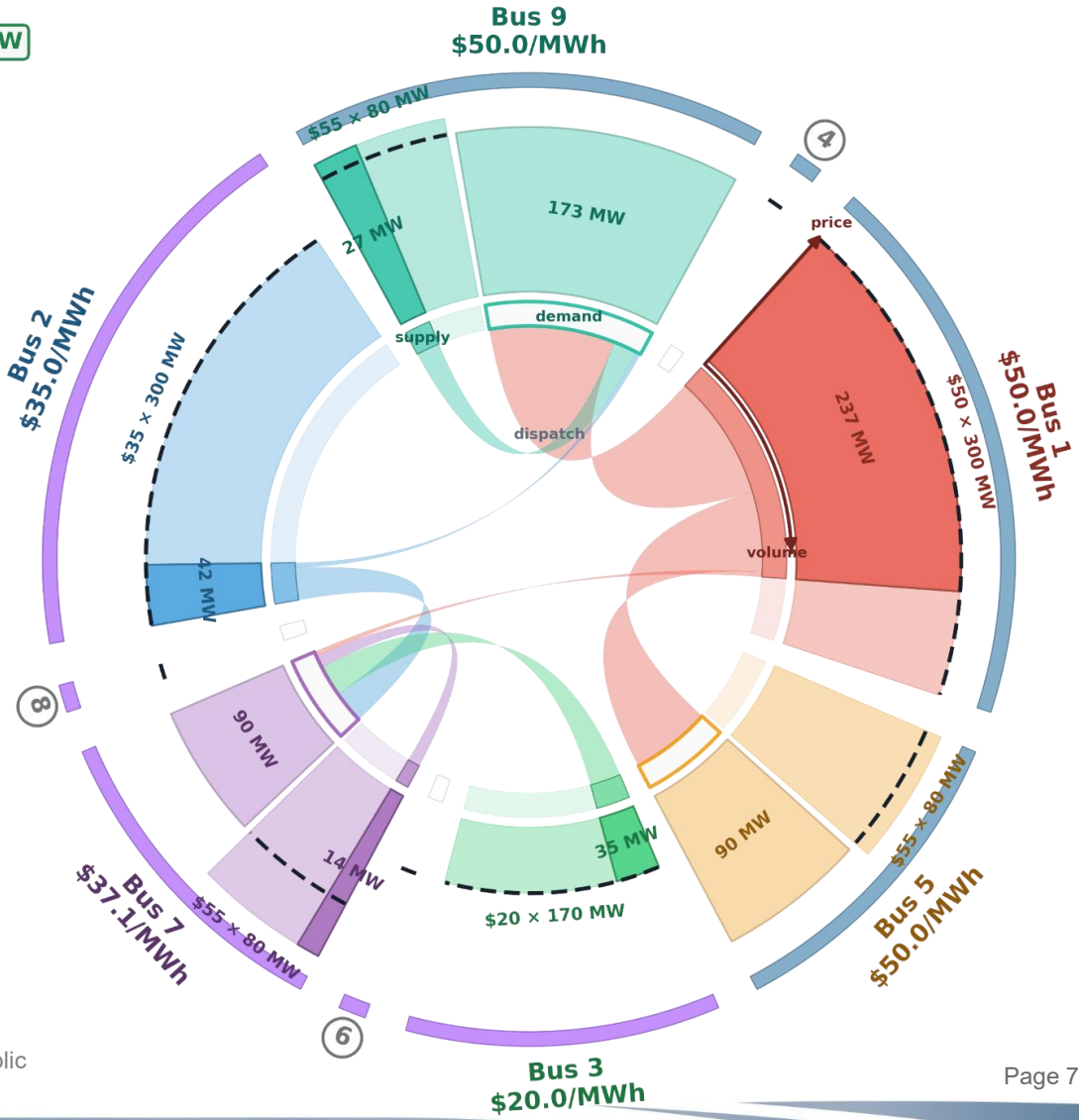
Example #3 - Calculation of relief

Tag	Priority	Scheduled (MW)	Impact (MW)	Share of relief	Relief (MW)	Curtailment or redispatch (MW)	Curtailed to (MW)
A1 (3→7)	7-F	25.0	21.3	34.6%	3.27	3.8	21.2
A GTL (net injections), untagged	7-F	64.9	18.7	30.4%	2.87	10.0	55.0
	gen at bus 2	49.1	5.2	8.4%	0.80	7.5	41.5
	gen at bus 3	15.9	13.5	21.9%	2.07	2.4	13.4
B2 (1→9)	7-F	173.1	21.6	35.1%	3.31	26.5	146.5
B1 (1→5, counterflow)	7-F	90.0	-12.2	not eligible	0.00	0.0	90.0
Total		**353.0**	**49.4**	**100.0%**	**9.45**	**40.3**	**312.7**



Example #3 - Resulting dispatch



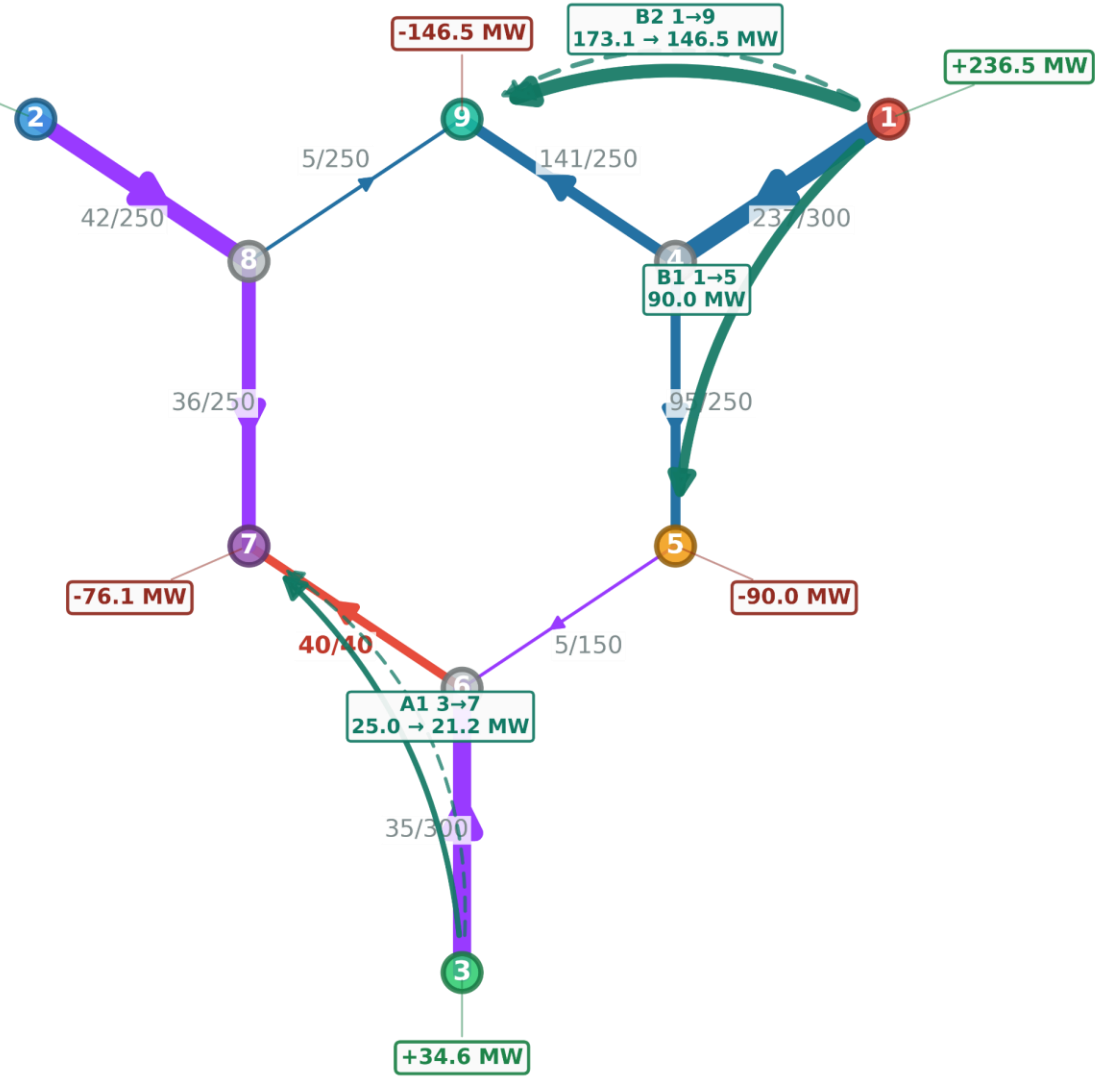


Example #4 – Relief through adjustment of transmission constraint in the market

- Start with Example #3 relief obligations
- Incorporate relief obligation as market flow constraint on the element
- ***Both markets dispatch to market flow constraints that reflect relief obligations***

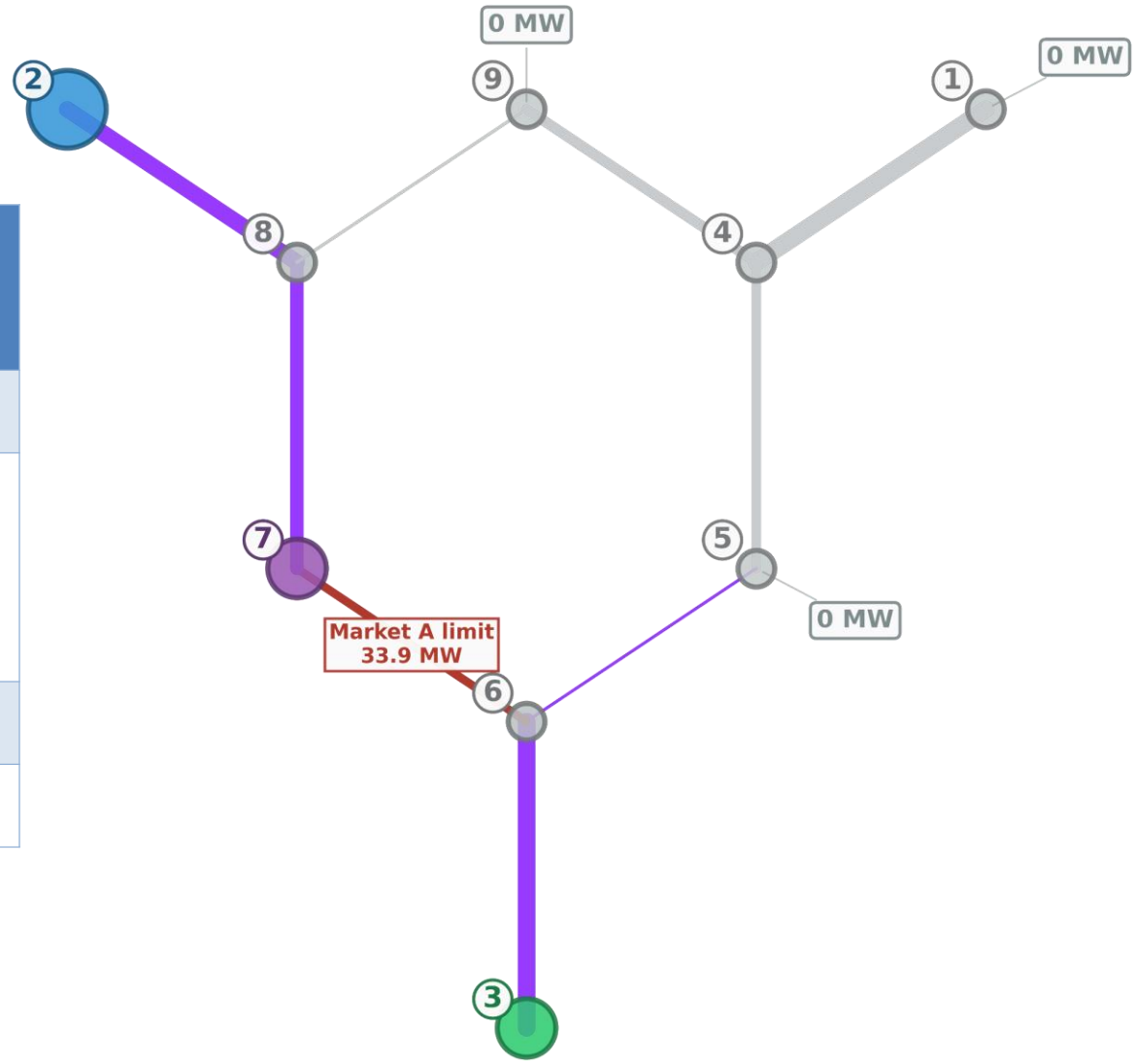
Starting with calculation of relief from example #3

Tag	Priority	Scheduled (MW)	Impact (MW)	Share of relief	Relief (MW)	Curtailment or redispatch (MW)	Curtailed to (MW)
A1 (3→7)	7-F	25.0	21.3	34.6%	3.27	3.8	21.2
A GTL (net injections), untagged	7-F	64.9	18.7	30.4%	2.87	10.0	55.0
	gen at bus 2	49.1	5.2	8.4%	0.80	7.5	41.5
	gen at bus 3	15.9	13.5	21.9%	2.07	2.4	13.4
B2 (1→9)	7-F	173.1	21.6	35.1%	3.31	26.5	146.5
B1 (1→5, counterflow)	7-F	90.0	-12.2	not eligible	0.00	0.0	90.0
Total		**353.0**	**49.4**	**100.0%**	**9.45**	**40.3**	**312.7**



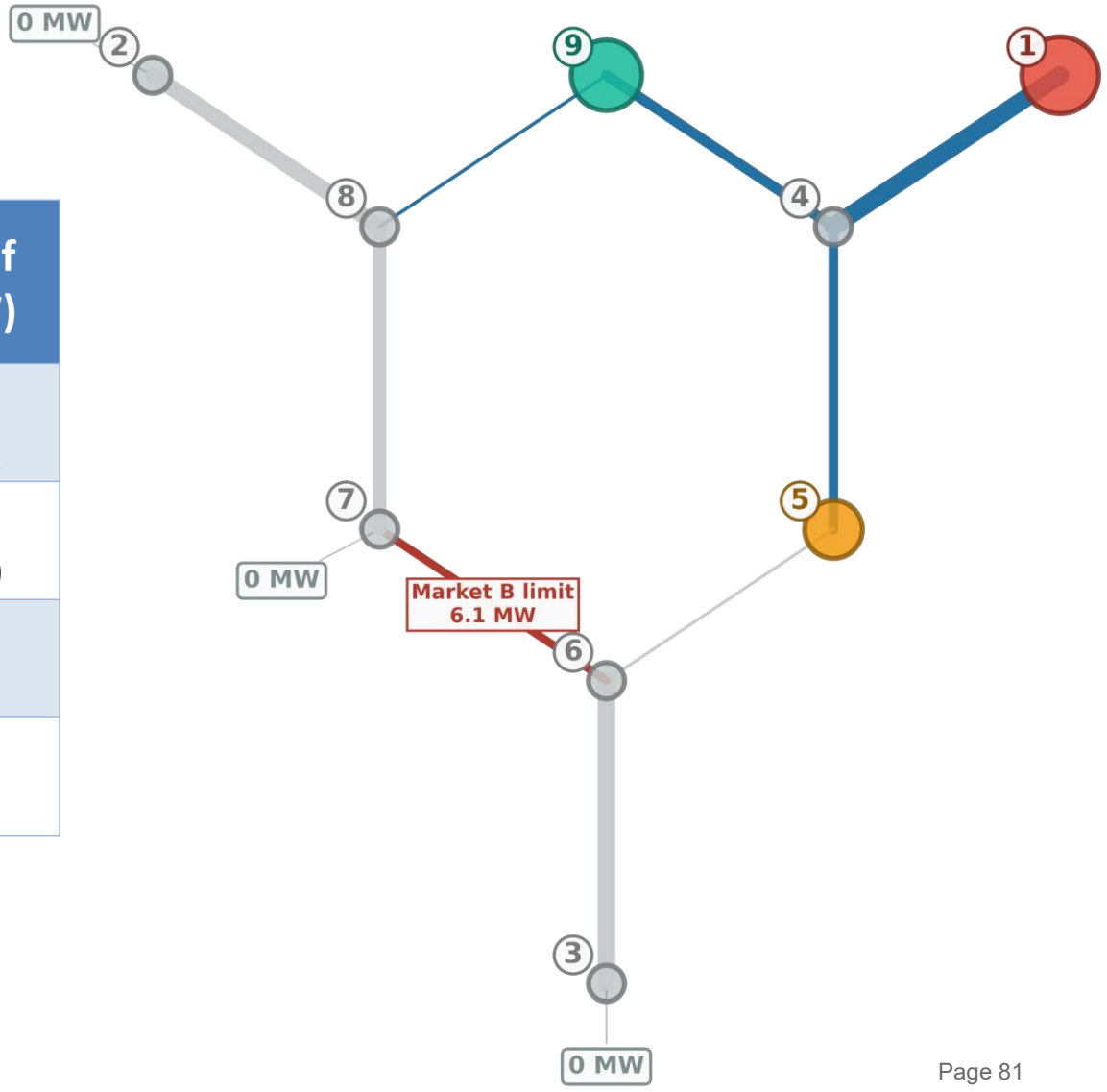
Market A - Calculation of market flow constraint based on relief obligation

Tag	Priority	Scheduled (MW)	Impact (MW)	Relief (MW)
A1 (3→7)	7-F	25.0	21.3	3.27
A GTL (net injections), untagged	7-F	64.9	18.7	2.87
Total		**89.9**	**40.0**	**6.1**
		Market Flow Limit	33.9	

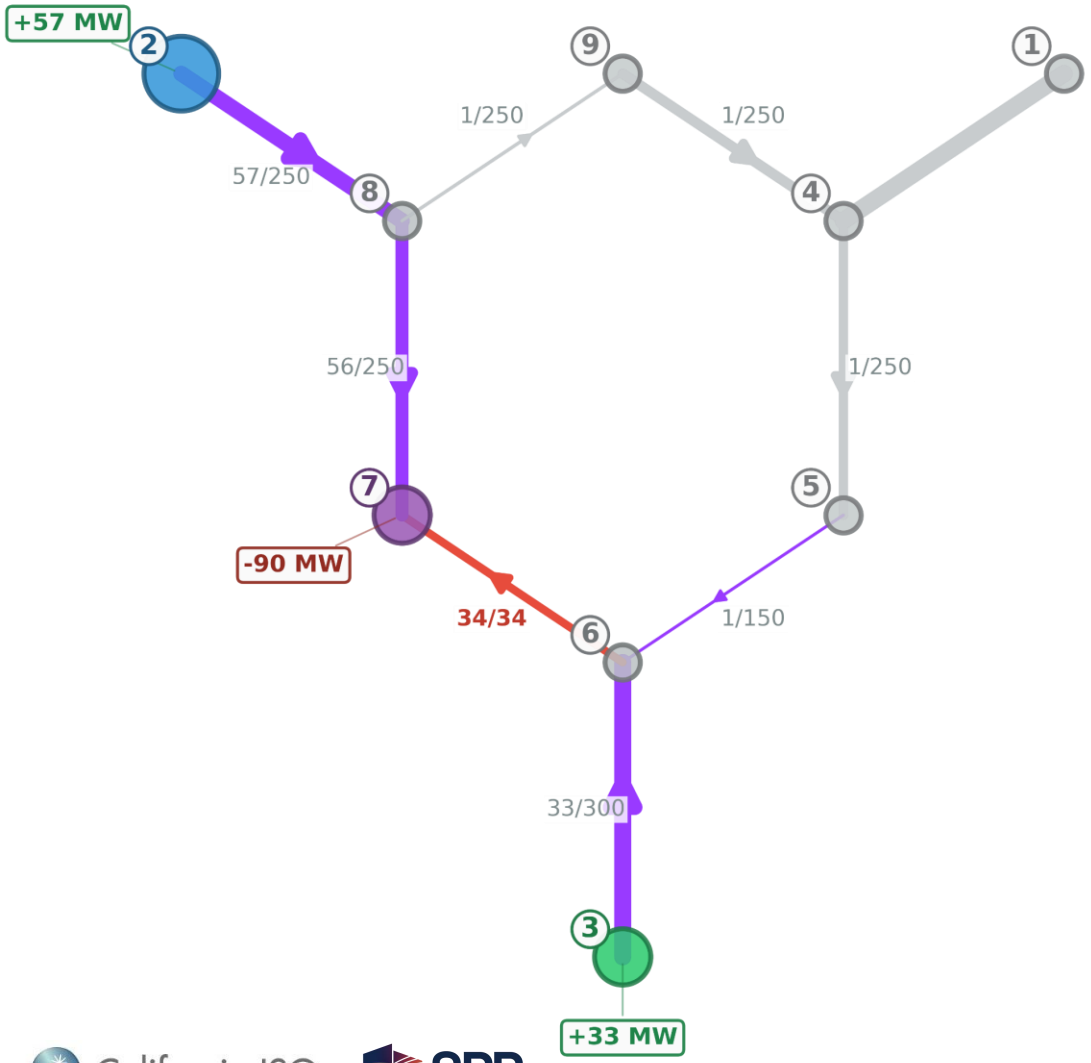


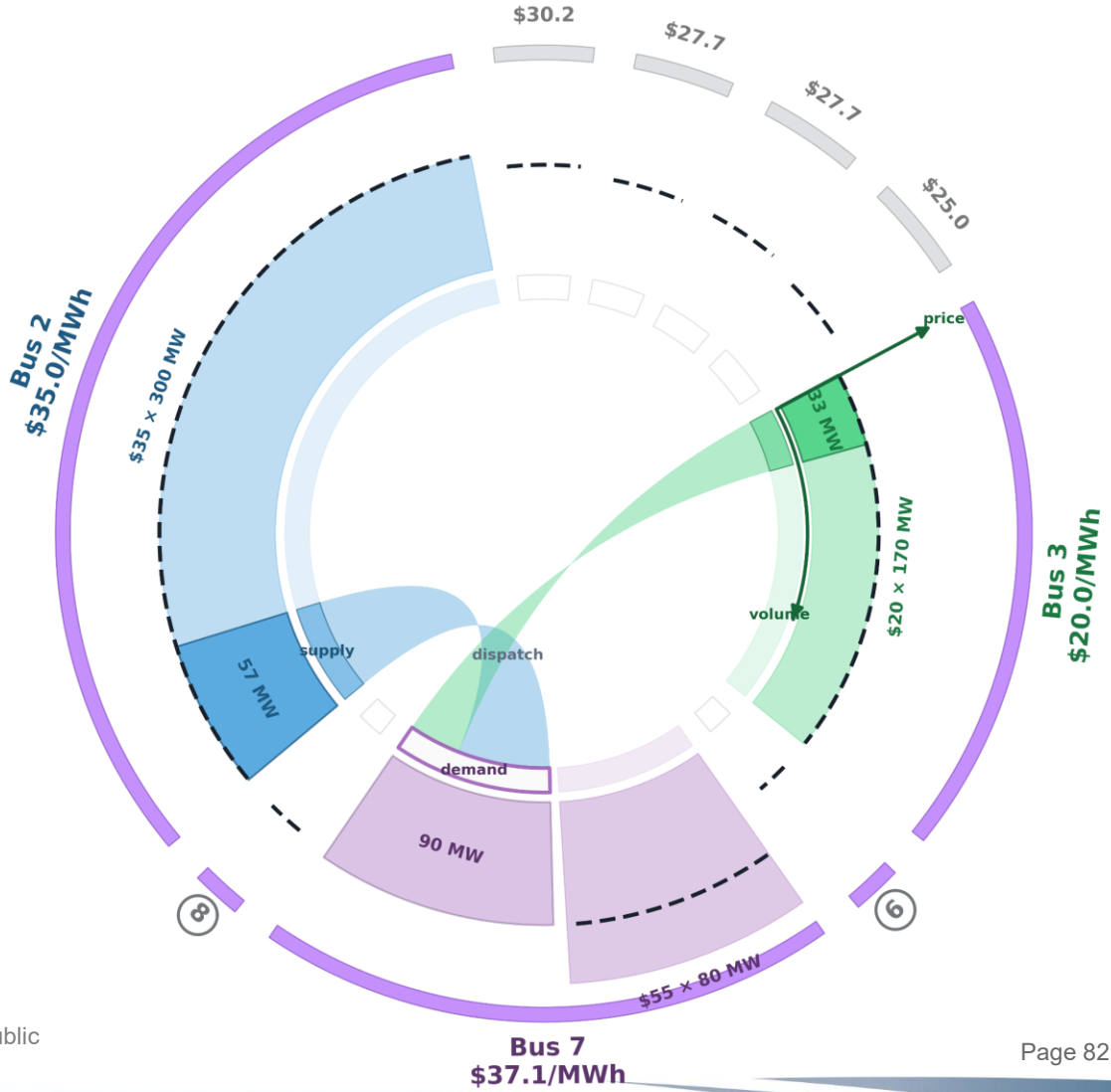
Market B - Calculation of market flow constraint based on relief obligation

Tag	Priority	Scheduled (MW)	Impact (MW)	Relief (MW)
B2 (1→9)	7-F	173.1	21.6	3.31
B1 (1→5, counterflow)	7-F	90.0	-12.2	0.00
Total		**263.1**	**9.4**	**3.3**
		Market Flow Limit	6.1	

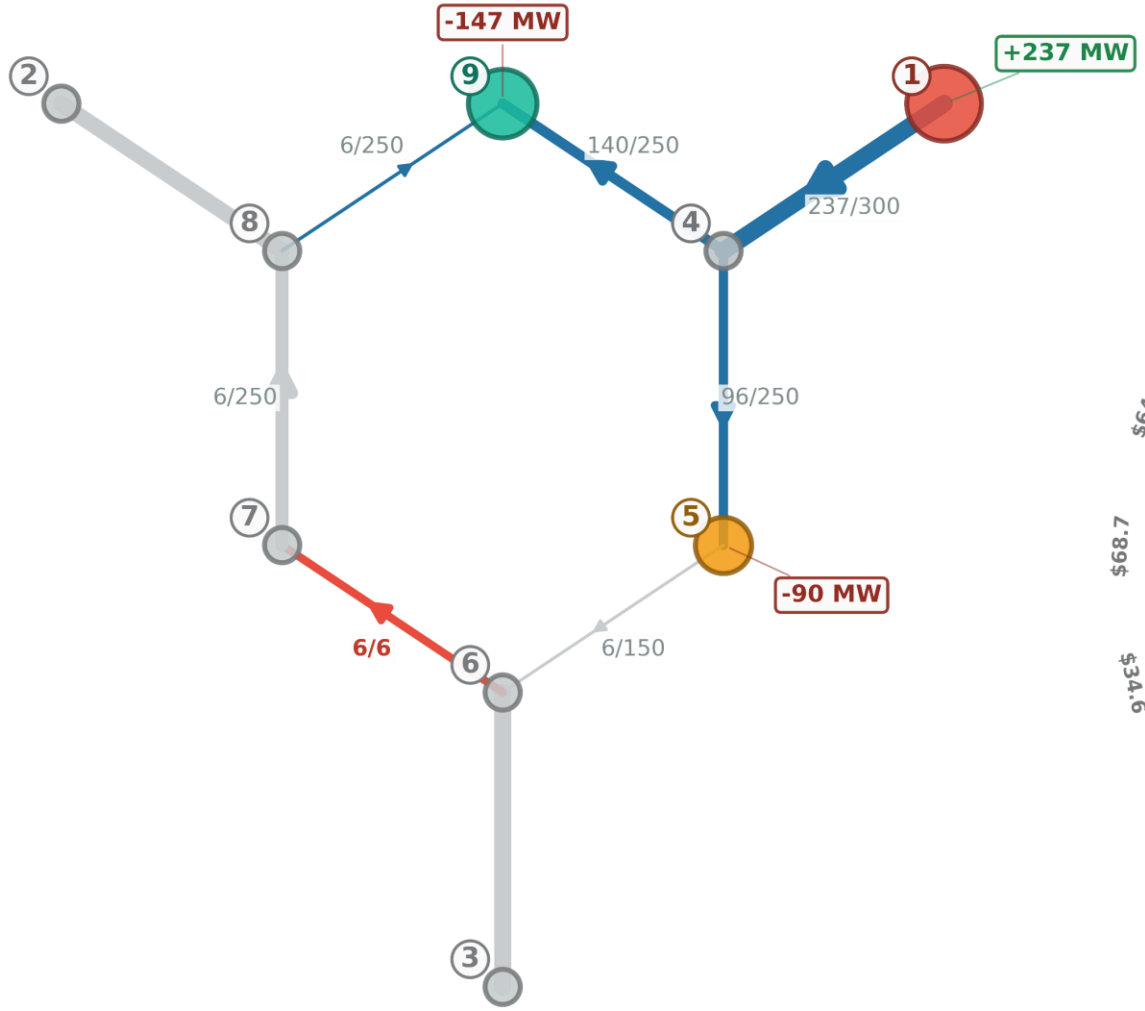


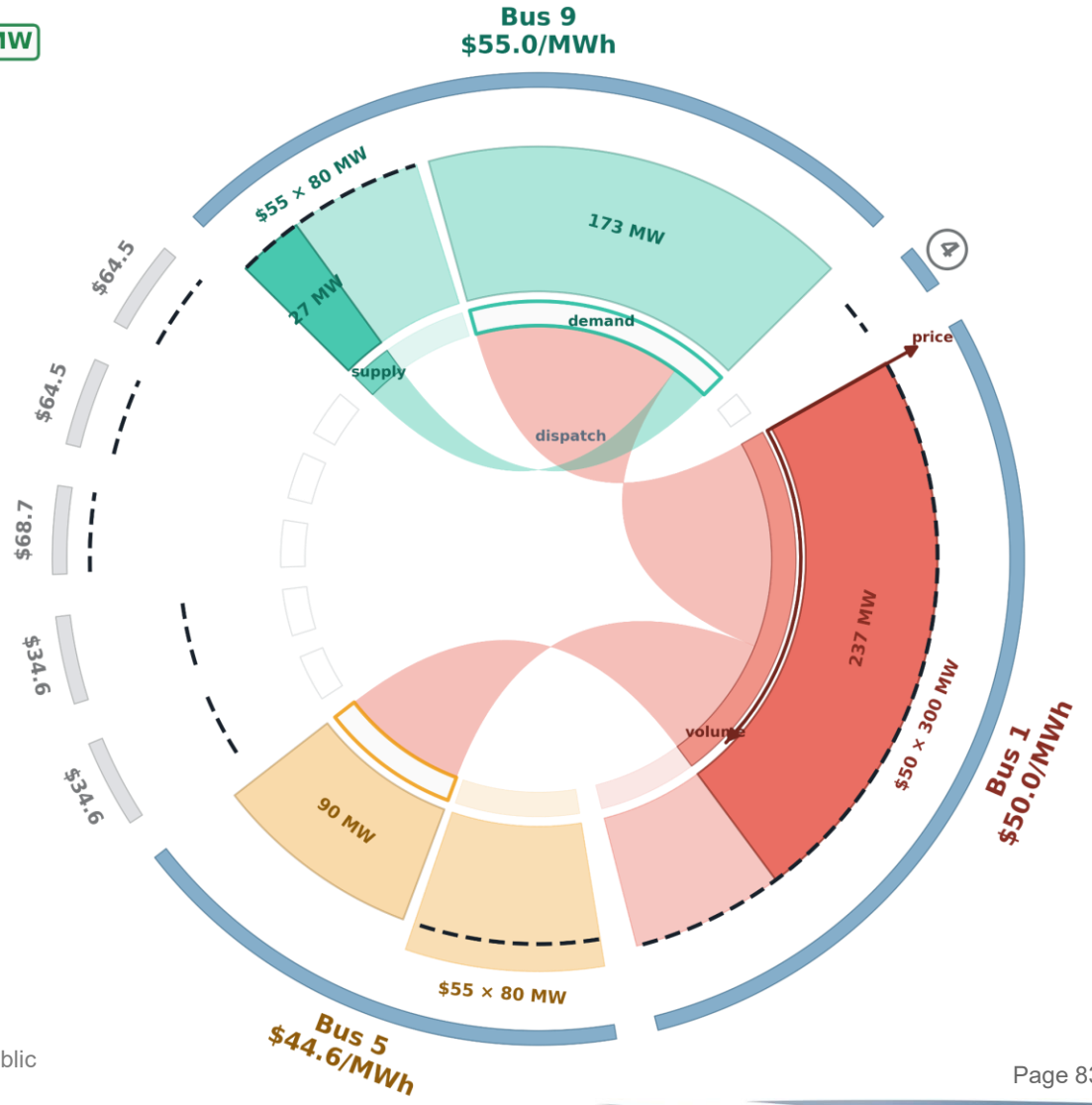
Example #4 – Market A dispatch with market flow limit



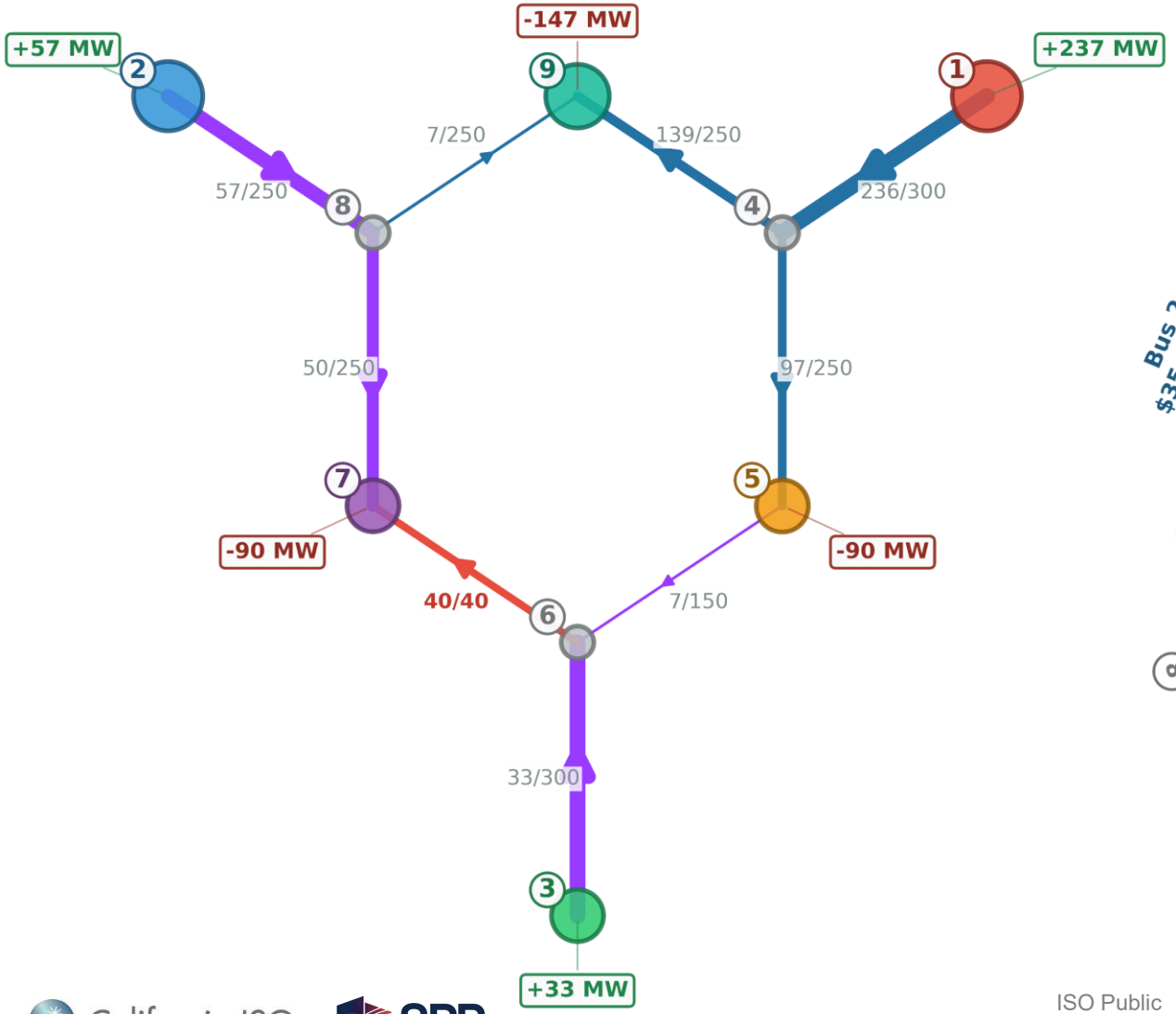


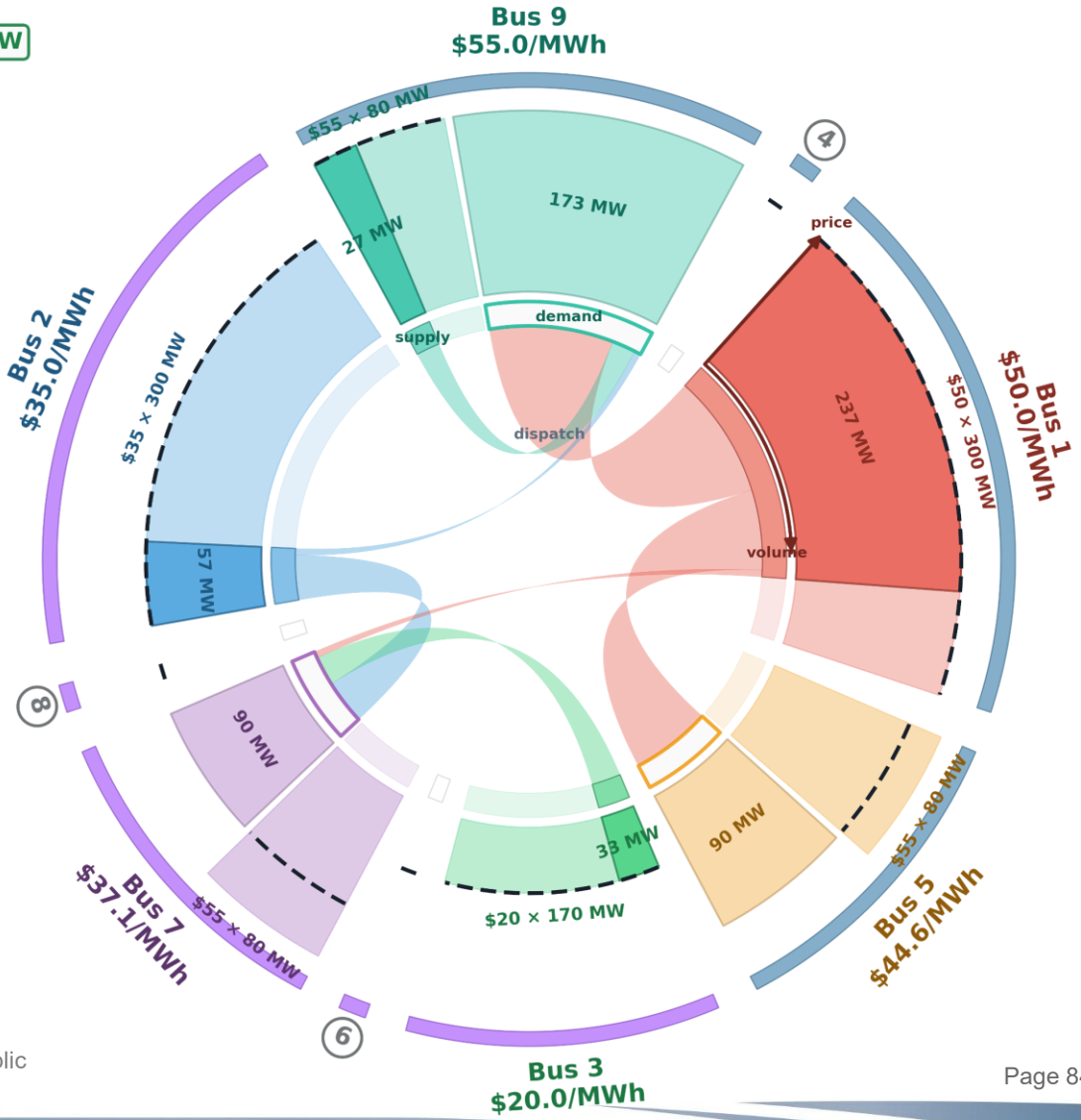
Example #4 – Market B dispatch with market flow limit





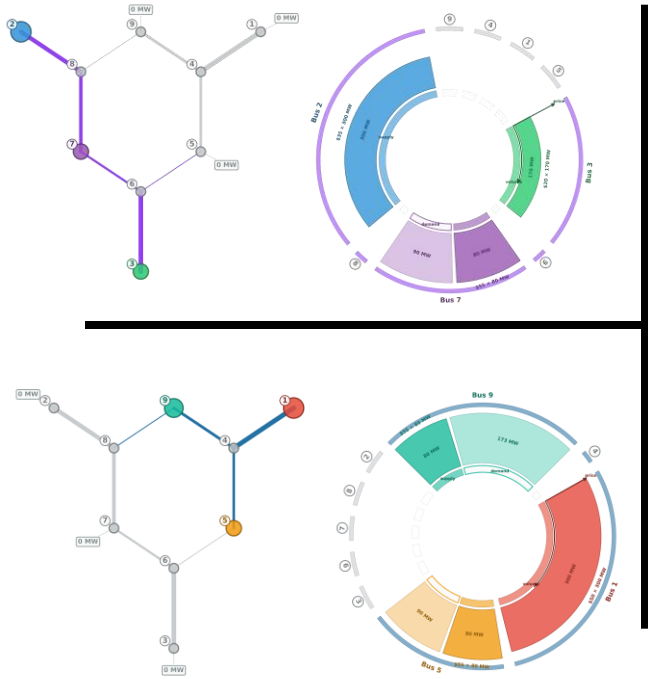
Example #4 – Network dispatch using market flow constraint



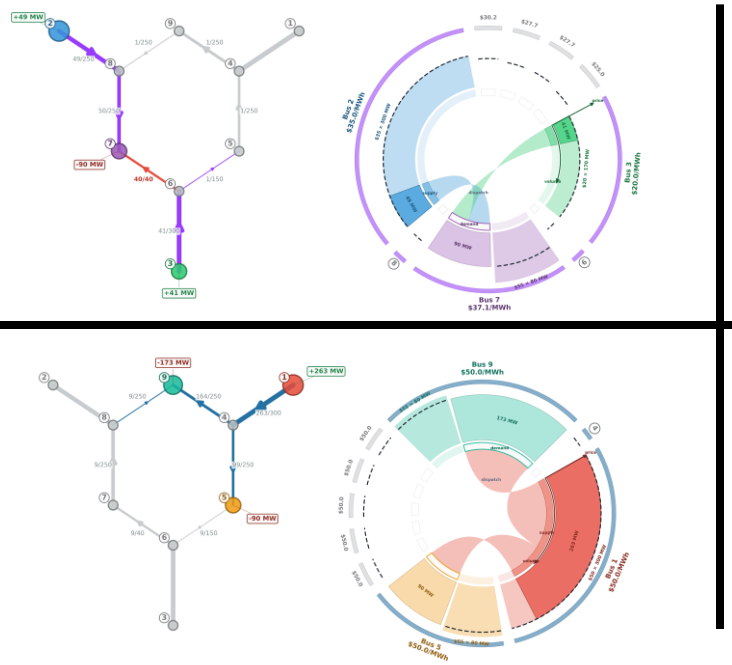


Coordination of parallel processes

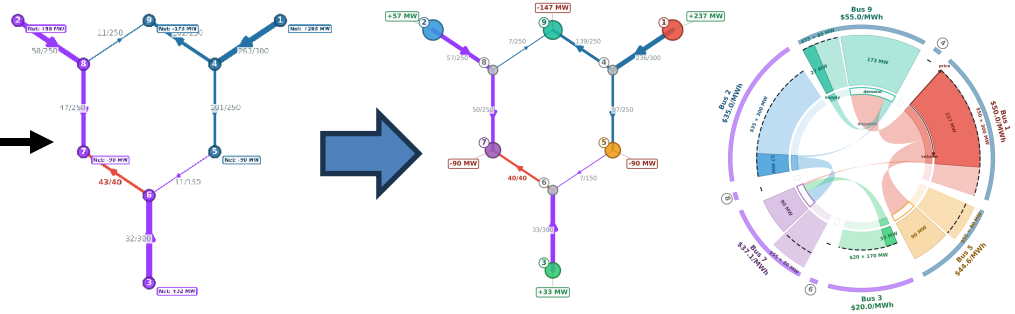
Inputs



Optimization



Reconciliation



Beginning of a framework for managing jointly impacted constraints

Developing a framework for managing jointly impacted constraints

- Eligibility criteria and process to identify constraints materially affected by multiple markets
- Methodology for determining relief obligations
- Agreement on methodology for providing relief through markets